The Effect of Oil News Shocks on Job Creation and Destruction

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Abstract

Using data from the Annual Survey of Manufactures (ASM) and the Census of Manufacturing (CMF), we construct quarterly measures of job creation and destruction by 3-digit NAICS industries spanning from 1980Q3-2016Q4. These long series allow us to address three questions regarding the effect of oil news shocks. What is the average effect of oil news shocks on sectoral labor reallocation? What characteristics explain the observed heterogeneity in the average responses across industries? Has the response of US manufacturing changed over time? We find evidence that oil news shocks exert only a moderate effect on total manufacturing net employment growth but lead to a significant increase in job reallocation. However, we find a high degree of heterogeneity in responses across industries. We then show that the cross-industry variation in the sensitivity of net employment growth and excess job reallocation to oil news shocks is related to differences in energy costs, the rate of energy to capital expenditures, and the share of mature firms in the industry. Finally, we illustrate how the dynamic response of sectoral job creation and destruction to oil news shocks has declined since the mid-2000s.

Keyword: job flows, oil news shocks, high frequency identification, OPEC announcements

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1 Introduction

Interest in understanding the effect of expectations about future oil prices on aggregate economic activity has been bolstered in recent years by geopolitical events and the plunge in global demand experienced during the COVID-19 pandemic. For instance, on 24 February 2022, Russia invaded Ukraine and significantly escalated the Russian-Ukranian War. As part of unified sanctions, the United States and many nations banned the import of Russian crude oil; this ban, if effective, would have stopped the approximately 700,000 barrels of Russian petroleum products imported into the US each day, a significant shock in national supply. Within two weeks, the price of the West Texas Intermediate (WTI) increased by more than 30%. Meanwhile, after falling by more than 16 million barrels per day in 2020, the Organization of the Petroleum Exporting Countries (OPEC) announced on 31 March 2022 that global oil demand was expected to grow to 100.9 million barrels per day in 2022, surpassing the pre-pandemic levels, and the monthly overall production level would increase by 0.432 mb/d in May 2022. How did the news about this future increase in production affect US job reallocation? More generally, how do oil price news affect the dynamic process of inter-firm job reallocation? Understanding how changes in expectations about future oil prices affect job gains and job losses is key to understanding business cycles and central to thinking about the effect of policies aimed at reducing the use of oil.

The key empirical challenge in estimating the effect of oil price shocks on economic activity is that these changes happen for a reason, and this reason might be endogenous to the US economy. The most common approach to deal with this endogeneity is to control for confounding variables in a structural vector autoregressive (SVAR) setup and rely on short-run or sign restrictions as an identification strategy. An alternative approach proposed by Känzig (2021) and inspired by the monetary policy literature is to focus on the change in the price of crude oil futures in a narrow window around OPEC meetings. That is the approach we follow here to investigate the effect of oil news shocks on sectoral job creation and destruction. Thus, we measure oil surprises from changes in the West Texas Intermediate (WTI) oil price futures around OPEC announcements and estimate the effect of oil news shocks on job reallocation using a proxy structural vector autoregressive model (Proxy-SVAR). Throughout this article, we refer to these shocks as oil news shocks instead of oil supply news shocks, as changes in crude oil futures prices may reflect OPEC's decision to modify

production targets to counteract the effect of a steep decline in demand for oil, as was the case during the COVID-19 pandemic.[1]

To assess the impact of oil news on sectoral job creation and destruction, we start by constructing time series of manufacturing job flows from confidential establishment-level data collected by the Census Bureau spanning the period between 1980:Q3 and 2016:Q4. More precisely, following Davis et al. (1998), we use quarterly data on the number of production workers reported in the Annual Survey of Manufactures (ASM) and the Census of Manufactures (CMF) to compute industry-level measures of job creation and destruction. The use of these confidential data enables us to construct longer and more disaggregated times series of manufacturing job gains and losses than publicly available from the Business Employment Dynamics. We then use a Proxy-VAR á la Känzig to estimate the response of sectoral job creation and destruction to oil news shocks.

We find that oil news shocks have a small negative and short-lived impact on manufacturing net employment growth, but a larger and longer-lasting impact on job reallocation. The news of a future increase in oil prices leads to a higher rate of job reallocation among sectors that are intensive in the use of energy (petroleum & coal products, wood products, and nonmetallic mineral products); in these sectors, the larger increase in job losses compared to gains results in lower employment growth. Perhaps not surprisingly, the largest declines in net employment growth and changes in job reallocation are experienced by the transportation equipment sector. In particular, expectations of higher crude oil prices result in a deeper and longer reduction in employment growth, as well as a more dynamic process of job reallocation for motor vehicle manufacturing than for any other industry. However, for other sectors that comprise a large share of manufacturing employment, such as food, we do not find a statistically significant response to news of higher oil prices. All in all, we find a considerable degree of cross-industry heterogeneity in the response to oil news shocks but, on average, we estimate responses that are smaller in magnitude than reported by earlier studies (e.g., Davis and Haltiwanger (2001), Herrera and Karaki (2015)).

We then turn to exploring what firm-level characteristics explain the differences in the magnitude of the response of job gains and losses and, in turn, net employment growth and excess job reallocation across industries. To do so, we exploit establishment-level data to compute job flows at a finer level of aggregation, 4-digit NAICS. We then construct measures of energy expenditures,

[1] See Degasperi (2023) and Kilian (2024).

capital expenditures, age, establishment growth, and other characteristics that may be correlated with the responsiveness of establishments' job flows to oil news shocks. We document that the sensitivity of job reallocation to oil news shocks is greater in industries that face higher energy costs, a higher ratio of capital to energy expenditures, and have a larger proportion of mature establishments.

What can we conclude from these findings? At first glance, the estimation results for aggregate manufacturing would seem to imply that oil news shocks cause rather small fluctuations in manufacturing jobs, despite their impact being statistically and economically significant for real GDP by Känzig (2021). However, a more careful look at the sectoral results reveals that focusing on aggregate data conceals the nature of the job reallocation dynamics triggered by an oil news shock. In particular, news of future oil price increases, whether arising from expectations of lower oil supply or increases in demand, alter the closeness of the match between actual and desired distribution of capital and labor inputs. Indeed, we find evidence that the shock triggers a more dynamic reallocation process in industries with higher energy costs and capital-to-energy expenditure ratios, such as wood, nonmetallic minerals, and paper. Moreover, our findings are consistent with the view that oil news shocks create anxiety among consumers, and, in turn, such concerns about higher future crude oil prices disrupt the demand for motor vehicles and related products (i.e., motor vehicle parts) leading to heightened job reshuffling in the transportation equipment sector. Given the importance of this sector in generating manufacturing employment (it accounted for 13.5% of manufacturing employment in the sample), it is crucial to understand the sensitivity of manufacturing job gains and losses to oil news shocks.

Finally, our estimation results reveal an additional empirical fact. We document that the response of sectoral job creation and destruction to an oil news shock is smaller in our sample than estimates obtained by previous studies using earlier samples. This discrepancy in the magnitude of the response could stem from three sources. First, a broad body of literature documents a decline in job market fluidity since the 1980s (see, e.g. Davis and Haltiwanger (2014)) linked to a decline in responsiveness to productivity shocks (Decker et al., 2018). Hence, the decline in the responsiveness of business-level employment to oil shocks could explain the smaller magnitude of our estimates. In the second place, important technological changes took place in the oil extraction process during our sample period. The shale revolution enabled the U.S. to significantly increase crude oil production

since the mid-2000s and become a net oil exporter in September 2019. Finally, contrary to other studies into the effects of oil price shocks on job reallocation, in this paper we identify shocks to future expectations of oil prices. Indeed, as our companion paper (Hanson, Herrera, and Zhao, 2025) reveals, responses to demand- and supply-driven shocks identified directly from a SVAR tend to be larger in magnitude. To explore whether the responsiveness of business-level employment has changed in recent years, we reestimate our Proxy-VAR on expanding sample windows that start from a subsample spanning 1980Q3-2000Q1, shortly after the US became a net oil exporter, and end with the full sample. This strategy allows us to assess whether job flows have become less responsive to oil news shocks over the years, while letting the relationship between the instrument and the oil price vary over time. In fact, we find striking evidence of a decline in the responsiveness of manufacturing job creation and destruction since the mid-2000s, which points to a change in the transmission of oil news shocks from the time of the shale boom.

This article is closely related to several investigations into the effect of oil price shocks on labor markets. First, our work contributes to a large body of literature that investigates the relationship between oil price shocks and the macroeconomy. In particular, we build on the work of Känzig (2021) who proposed the use of changes in oil futures prices around OPEC announcements to identify oil supply shocks. His work provides empirical evidence for a transmission channel of oil news shocks that operates through expectations of oil market conditions. We depart from Känzig (2021) in that we consider the identified shocks to be driven not only by the news of future changes in oil supply, but may also be driven by changes in demand (see Degasperi (2021)). Therefore, we refer to the shock as an oil news shock. More importantly, we leverage the confidential CMF-ASM data to document the response of job flows to oil news shocks and investigate whether cross-industry variation in the response to oil news shocks is systematically related to establishment characteristics.

Second, our work is related to investigations about the effects of oil price shocks on the destruction and creation of jobs in the US. Starting from the work of Davis and Haltiwanger (2001), several studies have shown that unexpected increases in oil prices lead to declines in employment growth and increased job reallocation (see, e.g. Davis and Haltiwanger (2001), Herrera and Karaki (2015), Karaki (2018)). This literature has underscored the importance of understanding the role that sectoral job gains and losses play as a mechanism whereby oil shocks result in economic recessions. However, these studies employ samples that end before the fracking revolution and identify oil price

shocks through short-term restrictions in SVAR models. Hence, their estimates could understate the role that the shale oil boom has played in how the labor market adjusts to oil news and rely on different identification assumptions.

Finally, this study complements the work of Bjørnland and Skretting (2024) who, using a time-varying parameter factor vector autoregressive model (TVP-FAVAR), found evidence of a change in the transmission mechanism of oil shocks since the shale boom. Indeed, they find that positive spillovers between oil and non-oil industries led to a change in the effect of oil supply shocks on US employment. More specifically, unexpected oil price increases –stemming from contractions in world oil supply– now give rise to employment growth in sectors that are connected to the oil industry. Their study is the first to provide evidence on how the dynamic response of investment, income, industrial production, and (non-oil) employment to oil supply shocks has evolved over a period of crucial technological change in the oil industry. However, because Bjørnland and Skretting (2024) examine employment rates at the industry and state level, their work is silent on the process of reallocation and restructuring of jobs among establishments that results from an oil news shock. Our study complements their work as we provide a richer characterization of the sectoral job creation and destruction dynamics, and we focus on the expectations channel. To investigate time-variation in the transmission of oil news shocks, we opt for a different estimation strategy than the TVP-FAVAR used by Bjørnland and Skretting (2024) for two reasons. Their TVP-FAVAR assumes the factors have the same loadings for all the sectors; instead by estimating each SVAR separately, we do not impose the assumption of common factors. In addition, our empirical strategy does not rely on short-run identification but on an IV strategy, and by using expanding windows we allow the relationship between the IV and the oil shock to vary over time. A drawback is that we do not exploit the richer data taken into account in the FAVAR setup.

The remainder of this article is organized as follows. Section 2 describes the data used in the empirical analysis, paying particular attention to the measures of sectoral job creation and destruction. Section 3 describes the empirical strategy. Section 4 discusses the dynamic response to oil news shocks, whereas Section 5 investigates the factors that account for the cross-industry heterogeneity in the responses. Section 6 investigates whether job flows have become less responsive to oil news shocks, and Section 7 concludes.

2 Data

In this paper, we explore the effect of oil news shocks on job reallocation using confidential data from the Annual Survey of Manufactures (ASM) and the Census of Manufactures (CMF)spanning from 1980 to 2016. In this section, we start by detailing the computation of job flows, compare these flows with publicly available measures of job gains and losses, and end with a description of the oil market data.[2]

2.1 Job Flows Data

Data on job flows for US manufacturing are calculated using establishment-level data collected by the Census Bureau. Specifically, we employ the quarterly number of production workers as reported at the establishment level in the ASM and the CMF. We supplement these data with total employment, openings/closings of establishments, and firm age from the Longitudinal Business Database (LBD). The LBD allows us to better evaluate whether an establishment that appears for the first/last time in the ASM is a true entry/exit in the market or whether it is the first/last time it appears in the surveys spanning our sample. Following Foster, Haltiwanger, and Kim (2006), we identify establishment births and deaths using the LBD age and initial year. Firms are only considered new births if the year of observation matches the initial year and the LBD age is one. This minimizes the risk of mislabeling a firm as a new birth if, in reality, it does not appear in the last sample but was sampled in the past.

We follow the methodology proposed by Davis and Haltiwanger (1998) to calculate job flows for total manufacturing, 20 three-digit subsectors of the North American Industrial Classification System (NAICS), and 4 four-digit NAICS industries in the transportation equipment sector. Letting EMP_{it} denote the employment in quarter t for establishment i, we define the growth of employment at the establishment level as:

$$g_{it} = \frac{(EMP_{it} - EMP_{it-1})}{Z_{it}}. \tag{1}$$

where

$$Z_{it} = 0.5 * (EMP_{it} + EMP_{it-1}). \tag{2}$$

[2]Because we compute job flows using quarterly establishment-level data, we aggregate the oil market variables to the same frequency.

This measure of employment growth has become standard in the literature on establishment dynamics because it accommodates entry and exit (see, e.g. Davis et al. (1998)). Then, we calculate the job creation and job destruction in group j (e.g., a three- or four-digit NAICS industry) as:

$$\begin{aligned} C_{jt} &= \sum_{i \in j^+} \Delta EMP_{it} \\ D_{jt} &= \sum_{i \in j^-} |\Delta EMP_{it}| \end{aligned}$$

where Δ denotes the first difference operator such that $\Delta X_t = X_t - X_{t-1}$. Thus, C_{jt} corresponds to the sum of employment gains in establishments that started or expanded in group j ($i \in j^+$) between quarters $t-1$ and t and D_{jt} corresponds to the sum of employment losses in establishments that contracted or closed in group j ($i \in j^-$) between quarters $t-1$ and t. The growth rates of job creation, POS_{jt}, and job destruction, NEG_{jt}, in industry j are then calculated as weighted sums of the growth rates at the establishment level a follows:

$$\begin{aligned} POS_{jt} = C_{jt}/Z_{jt} = \sum_{i \in j^+} (Z_{ijt}/Z_{jt})\, g_{ijt} \\ NEG_{jt} = D_{jt}/Z_{jt} = \sum_{i \in j^-} (Z_{ijt}/Z_{jt}) |g_{ijt}| \end{aligned}$$

Using these rates, we calculate the following job flows:

$$\begin{aligned} NET_{jt} &= POS_{jt} - NEG_{jt} \\ SUM_{jt} &= POS_{jt} + NEG_{jt} \\ EXC_{jt} &= SUM_{jt} - |NET_{jt}|. \end{aligned}$$

where the net change in employment growth, NET_{jt}, is given by the difference between job creation and job destruction, the gross reallocation, SUM_{jt}, can be interpreted as a size-weighted mean of the absolute value of the growth rates of the establishments, while the excess reallocation, EXC_{jt}, measures the amount of job movements in industry j above what is required to accommodate the observed net change in employment. All data series are seasonally adjusted using the Census Bureau's X-12-ARIMA seasonal adjustment program.

The first panel of Figure 1 shows the evolution of seasonally adjusted job creation and job destruction rates for total manufacturing along with the NBER recessions represented by gray-shaded areas. Note that these rates represent movements in the employment of production workers, since only data on these workers are reported on a quarterly basis. Job creation and job destruction rates in US manufacturing were more volatile in the early 1980s than in any other period in the sample. This increased volatility in the 1980s relative to later years is also evident in the job creation and destruction rates constructed by Foster et al. (2006) using a sample of the Longitudinal Research Database (LRD) that spans from 1972:2 to 1998:4.[3] As expected, job destruction rates largely exceeded job creation rates during recessions; pronounced and persistent spikes during economic downturns are evident for all recessions, but for the 1990-1991 recession. The figure also shows that U.S. manufacturing experienced a slow decline in job creation before and during the 2001 recession and the Great Recession. In addition, as noted in the literature (see, e.g. Davis and Haltiwanger (2014)), a decline in labor market fluidity is evident in the decreasing trend of job creation and job destruction rates since the late 1980s.

Figures 1 and 2 plot job creation and destruction for the three-digit NAICS industries and the 4 four-digit transportation sub-industries of interest. As the figures illustrate, two of the patterns observed at the aggregate level are also evident in the disaggregated data: the 1980s comprised a period of higher volatility than the later years, and job creation and job destruction exhibit a decreasing trend since the 1990s. This secular decline in the rates of job creation and job destruction is consistent with evidence found in many empirical studies regarding the decline of labor market fluidity since the late 1980s (see, e.g., Davis and Haltiwanger (2014)).

Table 1 reports summary statistics for job creation and destruction rates throughout the sample period (1980Q3-2016Q4). The mean is computed as the simple average in all quarters. The average quarterly job creation and destruction rates for manufacturing were 4.20 and 4.38 percent, respectively. Significant heterogeneity is evident across NAICS industries. For example, food manufacturing had the highest average creation and destruction rates, 6.10 and 5.90, while paper manufacturing had the lowest average job creation and destruction, 2.68 and 2.94. The table

[3]The LRD includes data on US manufacturing establishments collected in the Annual Survey of Manufactures (ASM) and the Census of Manufactures (CMF). The latter is a quinquennial census collected on years ending in '2' and '7' comprising the universe of manufacturing establishments. See Foster et al. (2006) for a detailed description of the sampling procedure of the ASM.

shows that, on average, the job destruction rate exceeded the job creation rate, leading to a slightly negative (-0.18) net employment growth in US manufacturing during the period. Several industries, such as food and kindred products, printing, wood products, and miscellaneous manufacturing, exhibited job creation rates that exceeded the destruction rates.

Table 1 also reports summary statistics for three additional job flow measures: NET, SUM, and EXC. As mentioned above, on average, net employment growth in manufacturing was slightly negative during the sample period. The rates of gross reallocation (SUM) and excess reallocation (EXC) in manufacturing –8.58 and 7.61, respectively– suggest a fluid labor market. In fact, while most sectors exhibited a slightly negative net employment change, the gross (excess) reallocation rate ranged from 5.62 (4.87) in paper to 12.00 (11.17) in food and kindred products. The fact that excess reallocation rates are sizable suggests that most reallocation occurred within establishments in the same industry. This fact, along with the reduction in net employment growth for most industries suggests that few jobs shifted from one industry to another.

2.2 Comparison of Job Creation and Job Destruction Rates

Our use of the ASM and CMF data to study the dynamic effect of oil price shocks has two advantages relative to the use of publicly available data from the Business Employment Dynamics (BED). First, the BED, generated from the Quarterly Census of Employment and Wages, covers all establishments subject to state unemployment insurance and comprises a broader set of establishments than the ASM/CMF. However, the BED data are not available prior to 1992 and hence would force us to exclude the 1980s from our study. Second, because the ASM contains establishment-level data on the use of inputs such as energy and capital, we are able to track which establishment-level characteristics matter for the transmission of oil news shocks.

However, the reader may wonder how the job creation and destruction rates computed from the ASM/CMF data compare with their BED counterparts. Thus, Figure 3 plots the job creation and destruction rates for the ASM/CMF and BED data. In general, the series tend to move together; the correlation between the two sources is 0.47 for job creation, 0.75 for job destruction, and 0.91 for net employment growth. The ASM/CMF rates exhibit slightly more variation and a higher mean. The standard deviation of job creation (job destruction) in the ASM/CMF is 0.68 (0.92) compared to 0.42 (0.84) in the BED. The average job creation and destruction rates are 4.10 and

3.92 in the ASM/CMF data versus 0.92 and 0.84 in the BED.[4] On average, measures computed from the ASM/CMF are about 0.2 percentage points higher than their BED counterparts.

Earlier work by Davis and Haltiwanger (2001) used data from the Longitudinal Research Database (LRD) spanning 1972:2 to 1988:4 to study the response of sectoral job creation and destruction to oil price changes. Because the data sources are the same (ASM/CMF) and the object of study is closely related, we also compare our job creation and destruction rates with those reported by Davis and Haltiwanger (2001). Table 2 reports job creation and destruction rates for our 1980Q3-2016Q4 sample, the BED sample spanning 1992Q2 to 2016Q4, and Davis and Haltiwanger (2001) LRD rates covering the earlier period between 1972Q2 and 1988Q4. As the table illustrates, the average rates for 1972-1988 exceed the rates we computed for the later sample by about one percentage point. Smaller rates of job creation, and especially job destruction, in the 1990s compared to earlier years are also documented by Foster et al. (2006) who report an average job creation (destruction) rate of 4.98 (4.41) for 1994Q1-1998Q4 versus a 5.07 (5.47) for 1972Q2-1993Q4. Lastly, an interesting similarity between most samples and sources is that job destruction exceeds job creation, which reflects the large spikes in job destruction rates during employment downturns. The only exception is the 1994Q1-1998Q4 period reported by Foster et al. (2006) when the US economy expanded.

2.3 Oil Market Data

The oil market variables considered in this paper follow the baseline specification of Känzig (2021). That is, the oil market block in the vector autoregressive (VAR) model contains the real price of oil, world oil production, a measure of oil inventories, and an index of real global economic activity. Unless otherwise specified, oil market variables are measured in logarithms.

As a measure of real oil prices, we employ the spot price of the West Texas Intermediate (WTI) deflated by the Consumer Price Index (CPI). Data for both variables are obtained from the Federal Reserve Economic Data (FRED) maintained by the Federal Reserve Bank of St. Louis.

Data on world oil production (including lease condensate production) are obtained from the Energy Information Administration (EIA). Oil production is measured in thousands of barrels per

[4]Similarly, Foster et al. (2006) find the BED job creation and destruction rates to be more volatile than their LRD counterparts over 1972-1998.

day. To construct a measure of crude oil inventories, we follow the literature and obtain data on total crude oil inventories and petroleum stocks for the US, as well as data for OECD petroleum stocks from the Energy Information Agency (EIA). Then, following Hamilton (2009) and Kilian and Vigfusson (2014), we calculate world inventories by scaling the total US crude oil inventories by the ratio of OECD petroleum stocks to U.S. petroleum stocks.

As mentioned above, our baseline specification follows Känzig (2021). Therefore, to measure global economic activity, we aggregate industrial production from the OECD, Brazil, China, India, Indonesia, the Russian Federation, and South Africa. This index of world industrial production comprises real production in the manufacturing, mining, electric and gas industries.[5] As we will show in Section A.1, our estimation results are robust to replacing the world industrial production index with Kilian (2009)'s real economic activity index.[6]

To identify the effect of oil news on job flows, we use a quarterly measure of oil news calculated by exploiting the variation in oil price futures around announcements from the Organization of Petroleum Exporting Countries (OPEC) as in Känzig (2021). In order to be able to interpret the change in oil price futures as stemming from the announcement and not from other information regarding other variables in the system (e.g., changes in global economic activity), a tight window around the announcement is used. Therefore, oil surprises are computed as follows. Let $F^h_{m,d}$ denote the futures price for the West Texas Intermediate (WTI) h-month contract on day d of month m, then the oil surprise $OS^h_{m,d}$ is calculated as the logarithmic difference between the price of the WTI oil futures on the day of the announcement and the day before the announcement as:

$$OS^h_{m,d} = log(F^h_{m,d}) - log(F^h_{m,d-1}). \tag{3}$$

To take into account implementation delays and the horizon spanning OPEC announcements, Känzig uses futures contracts with maturities ranging from one month to a year to calculate the surprises. The first principal component of these surprises is then used as a measure of oil supply news. Here, we follow the same practice and then aggregate all the surprises in a quarter to obtain a measure that matches the frequency of the job flows data. If there is no announcement in a

[5]The index is available at: https://sites.google.com/site/cjsbaumeister/datasets.

[6]This index is available from the St. Louis Federal Reserve Bank Economic Data (FRED) at https://fred.stlouisfed.org/series/IGREA.

quarter, then the quarterly surprise is set to zero. The series of quarterly news is then used as an instrument for oil price changes in a Proxy-VAR.

Figure 4 shows the evolution of the quarterly oil news series over time. Larger spikes, such as in August 1986 or after September 11, 2001, represent instances where the announcement led to a large revision of oil price expectations. While Känzig refers to this variable as a measure of oil supply news, we refer to these shocks as *oil news shocks* given that OPEC decisions to alter production quotas could stem from agents' beliefs about the path of future demand (see, e.g., Kilian and Zhou (2023) and Alsalman et al. (2023)).

3 Empirical Strategy

To study the effect of a shock to oil price expectations (an oil news shock hereafter), we first estimate a Proxy-VAR model (Stock and Watson (2012) and Mertens and Ravn (2013)) à la Känzig (2021) where oil surprises are employed as an instrument. That is, we use the oil surprise series derived from the daily oil futures data, aggregated to a quarterly frequency (see Section 2.3), as an instrument to identify the oil news shock.[7]

Hence, we start by estimating a reduced-form VAR(p) given by

$$\boldsymbol{y_t} = \boldsymbol{c} + \boldsymbol{B_0} + \boldsymbol{B_1} y_{t-1} + ... + \boldsymbol{B_k} y_{t-p} + \boldsymbol{u_t} \tag{4}$$

where $\boldsymbol{y_t}$ is a 6×1 vector that contains the oil market variables followed by the job creation and destruction rates of interest (e.g., the job creation and job destruction rates for total manufacturing or for one of the 3-digit or 4-digit NAICS industries); $\boldsymbol{c}$ is a 6×1 vector of intercepts; $\boldsymbol{B_i}$ are 6×5 matrices of autoregressive coefficients; and $\boldsymbol{u_t}$ denotes a 6×1 vector of Gaussian innovations with mean $\mathbf{0}$ and variance $\boldsymbol{\Sigma}$. The oil market variables include the real price of oil, world oil production, world oil inventories, and a measure of real economic activity, in that order. We rotate each pair of job creation and job destruction rates (one pair for the aggregate or the industry of interest), after the oil variables. We set the lag order p to 4 and estimate the model in log levels.[8]

[7] Proxy-VARs (SVAR-IV) have also been commonly used in the literature to estimate the effect of monetary policy and uncertainty shocks. See, among others, Carriero et al. (2015), Gertler and Karadi (2015), Caldara and Kamps (2017), and Stock and Watson (2018)).

[8] Estimation results reported in section A.1 show that our findings are robust to alternative specifications.

As is common in the literature, we assume that the reduced-form innovations, $\boldsymbol{u_t}$, are related with the structural shocks, $\boldsymbol{\epsilon_t}$ through a linear mapping

$$\boldsymbol{u_t} = \boldsymbol{S}\boldsymbol{\epsilon_t} \tag{5}$$

where $\boldsymbol{S}$ is a non-singular 6×6 matrix that captures the impact effect of the structural shocks. The structural shocks are assumed to be i.i.d. and mutually uncorrelated with mean $\mathbf{0}$ and variance Ω. Given the linear mapping in equation (5), and because we are only interested in identifying the structural effect of a shock to the real oil price, all we need to do is identify the first column of $\boldsymbol{S}$, which we denote by $\boldsymbol{s_1}$.

Then, to identify the structural impact vector we use an external instruments approach and employ the quarterly oil surprise series, z_t, as an instrument. Recall that for the oil news series to be a valid instrument, it is required that:

$$\mathbb{E}[z_t \boldsymbol{\epsilon_t}] = \alpha \neq 0 \tag{6}$$

$$\mathbb{E}[z_t \boldsymbol{\epsilon_{2:6,t}}] = 0 \tag{7}$$

where $\epsilon_{1,t}$ denotes the oil news shock and $\epsilon_{2:n,t}$ denotes the 5×1 vector containing the other structural shocks in the system. The reader will recognize equation (6) as the relevance assumption and equation (7) as the exogeneity assumption, which allows us to identify the vector s_1 (up to sign and scale) as

$$\bar{s}_{2:6,1} \equiv s_{2:6,1}/s_{1,1} = \mathbb{E}[z_t u_{2:n,t}]/\mathbb{E}[z_t u_{1,t}] \tag{8}$$

as long as $\mathbb{E}[z_t u_{1,t}] \neq 0$. This yields the structural impact vector $s_1 = (s_{1,1}, \bar{s}_{2,1} s'_{1,1})'$ where the scale of $s_{1,1}$ is normalized such that $\Sigma = S\Omega S'$. For ease of interpretation, we normalize the oil news shock such that it corresponds to a 10 percent increase in the price of oil by setting $\Omega = diag(\sigma^2_{\epsilon_1}, ..., \sigma^2_{\epsilon_6})$ and $x = s_{1,1}$.[9]

[9]For more detailed analysis of the assumptions, theory, and proof, see Mertens and Ravn (2013) and the appendix of Känzig (2021).

4 The Effect of Oil News on Job Reallocation

This section first describes the baseline Proxy-VAR estimation results for the oil market variables and aggregate manufacturing responses. It then discusses the estimates for the three-digit NAICS industries, and follows by exploring the responses of 4 four-digit subindustries in the transportation sector.

4.1 Oil market and aggregate manufacturing responses

We begin this section by describing the estimation results for the oil market block. Before we discuss our results, it is important to note that because the job creation and destruction data start in 1980, our sample excludes most of the years when the oil futures market did not exist or transaction volumes were low. In contrast, Känzig (2021) uses monthly data starting in 1975. The exclusion of the 1970s might be seen as a drawback, since the great variability of oil prices during this period could help to better identify the effect of an oil price shock. However, excluding this part of the sample has some advantages when identifying oil news shocks using a proxy variable constructed from daily futures data. In particular, as Kilian (2024) notes, oil futures only started trading in 1983, longer maturities were introduced later, and some periods saw no or low trading. Hence, the extracted principal component may be affected by data censoring.

As is common in the literature, all responses are normalized so that an oil news shock leads to a 10 percent increase in the real price of oil. As Figure 5 illustrates, an oil news shock leads to an immediate increase in the oil price which continues to be above the historical average for about five years. World oil inventories exhibit a slight decrease on impact, but an increase in the long run. Oil production responds positively in the first quarters, although not in a statistically significant manner, which is consistent with oil producers responding to expectations of higher future oil prices. Furthermore, the transitory increase in world oil production suggests that oil news shocks convey beliefs of heightened aggregate demand (Kilian and Zhou (2023) and Herrera et al. (2023)) and do not only represent expectations of oil supply cuts (Känzig (2021)).

We now turn our attention to the response of US manufacturing job flows to oil news shocks. The top left panel of Figure 6 depicts the response of US manufacturing job creation (solid red line) and job destruction (blue dashed line) to an oil news shock that results in a 10% increase in

the real price of oil. The 68% confidence bands, denoted by shaded areas, are computed using a moving block bootstrap (see Jentsch and Lunsford (2019)) with 10,000 bootstrap replications. In addition, the figure plots the response of the net employment (NET) growth (black dotted line).

The estimated impulse responses reveal a small increase in job creation on impact that becomes statistically significant two quarters after the shock. With no initial increase in job destruction, net employment growth increases moderately. However, job destruction rises significantly three to four quarters after the shock, leading to a reduction in net employment growth. This decline in employment growth is rather short-lived, as evidenced by the close to zero (0.02 percentage points) fourth-quarter cumulative change (see the second column of Table 3). Now, while the fourth-quarter cumulative increase in net employment growth is negligible, news of an increase in oil prices leads to a statistically significant increase in job reallocation of 0.54 percentage points (about 6% of the average quarterly rate). Two years after the shock, both job creation and destruction have largely returned to their pre-shock levels.

In brief, our estimation results indicate that oil news shocks have a small negative and short-lived impact on net employment growth. This contrasts with the findings of previous studies such as Davis and Haltiwanger (2001) and Herrera and Karaki (2015) where oil price shocks are estimated to result in a significant reduction in employment growth that lasts over a year. There are several potential causes for these differences. First, while the above-mentioned studies used SVARs where the oil price shock was identified via short-run restrictions, we use a Proxy-VAR that identifies unexpected changes in oil prices related to OPEC news. Hence, while the first identification scheme captures movements in oil prices that may be due to supply, demand, or precautionary demand motives, the latter measures the effect of the surprise component of OPEC policy decisions. Second, due to data availability, our sample excludes the years prior to 1980 which are included in previous studies but includes data for the 1990s and 2000s. As we will discuss in detail in Section 6, had we used a shorter sample, we would have estimated a larger impact of oil news shocks on the reallocation of US manufacturing jobs.

4.2 Sectoral Job Flows

Figure 6 reports the estimated impulse response functions for the nondurable manufacturing industries. A considerable degree of heterogeneity is evident in the response to the news of an

oil price increase. First, industries such as food products, textiles & fabrics, textile mill products, apparel, petroleum & coal, and paper experience significant variation in job creation and destruction during the first year after the shock. For all these industries, the increase in job destruction exceeds the increase in job creation, leading to declines in net employment change around the second quarter after the shock. However, as Table 3 reveals, four quarters after the shock, the cumulative changes in net employment growth are statistically insignificant, indicating that the response is short-lived. However, as is the case for total manufacturing, the insignificant response of net employment growth masks significant changes in reallocation at the establishment level. As Table 3 illustrates, excess job reallocation (the amount of job movement above what is required to accommodate the change in employment) decreases on impact for leather products, printing, and rubber & plastics, whereas the fourth-quarter cumulative change in excess reallocation is positive for paper and petroleum & coal. The 0.67 percentage point cumulative increase in excess reallocation in petroleum and coal is particularly notable, as given the historical average excess reallocation rate of 5.85%, this represents approximately an 11% increase in job movements needed beyond what is required to accommodate the 0.08 percentage point drop in net employment growth. This increase constitutes the largest change in EXC among nondurable industries.

The responses for durable industries are reported in Figure 7. As was the case for nondurable industries, we observe considerable heterogeneity in the impulse responses. Significant increases in job creation and destruction are estimated for wood products, nonmetallic mineral products, furniture, and transportation equipment in the first year after the shock. For some of these industries, the initial increase in job destruction exceeds the rise in job creation (e.g., nonmetallic mineral products, transportation equipment), thus net employment growth falls on impact; only a couple of three-digit industries (e.g. wood products, furniture) experience increases in net employment growth on impact (see Table 3). The effect of the oil news shock on net employment growth is rather short-lived. A year after the shock, the cumulative effect is positive for six out of the 10 three-digit industries; economically significant declines in net employment growth are observed for nonmetallic mineral products (-0.34) and transportation equipment (-0.38), although they are estimated with a low degree of precision.

As Table 3 illustrates, a year after the oil news shock the process of firm entry, exit, growth, and contraction has intensified for about half of industries. Statistically significant increases in job

reallocation are estimated for nonmetallic mineral products (1.03 percentage points), transportation equipment (1.05), and furniture (0.51). More than half of the job reallocation in these industries is needed to accommodate the drop in net employment growth. This is evidenced by the magnitude of the cumulative change in excess reallocation four quarters after the shock, which ranges between 0.39 for furniture and 0.69 percentage points for nonmetallic mineral products. That industries such as wood and nonmetallic mineral products exhibit large changes in job reallocation is perhaps not surprising. After all, wood production (i.e., the forest product subsector) is the second largest consumer of fuels and power in US manufacturing, with approximately 75% of fuel needs being satisfied with distillate fuel and oil (Department of Energy (2013)). This is in part due to the heavy use of steam and extreme heat in producing the products. Similarly, the industry of nonmetallic mineral products relies heavily on heat to refine minerals, which is an energy intensive and costly process.

Three takeaways are derived from the estimation results presented in this section. First, there is a great degree of heterogeneity in the response of sectoral job flows to oil news shocks. Sectors that have traditionally been considered to be intensive in the use of oil (petroleum & coal products, wood, and nonmetallic mineral products) experience contractions in employment and increased labor reallocation. Second, the effect of an oil news shock that results in a 10% increase in the price of crude oil on sectoral job creation and destruction is shorter-lived, yet it leads to substantial job reallocation in several nondurable and durable industries. These findings are consistent with empirical and theoretical literature that suggest that oil price shocks lead to reallocation of workers from industries that heavily rely on oil as an input for manufacturing or consumption (such as transportation equipment) to less energy-intensive industries. Finally, our results indicate that the transportation sector continues to play an important role in the transmission of oil price shocks. In the following section, we further inquire into the effect of oil news shocks on this sector.

4.3 Digging deeper into the response of the transportation sector

The results presented in the previous section suggest that the transportation equipment sector plays a key role in the dynamic adjustment of manufacturing jobs after an oil news shock. To further investigate the labor reallocation responses in this sector, we re-estimate our model for 4 four-digit sub-industries: motor vehicles, motor vehicle bodies and trailers, and motor vehicle parts

manufacturing. The motivation for such a decomposition is threefold. First, the transportation equipment industry includes all forms of transportation equipment such as boats and airplanes, which may respond differently to expectations of higher oil prices than motor vehicles. Second, work by Ramey and Vine (2011) suggests that despite the reduced impact of oil price changes on the US economy since the 2000s, motor vehicle consumption continued to be quite sensitive to fluctuations in oil prices. Finally, as Table 1 illustrates, transportation equipment represented 13.46% of manufacturing employment between 1980Q3 and 2016Q4, the largest share among all three-digit industries. In addition, among the subindustries in this sector, motor vehicles and motor vehicle bodies exhibited higher average job creation and destruction rates than most manufacturing industries. In brief, we consider the importance of digging deeper into the responses of the transportation sector in order to better understand how oil news shocks affect labor dynamics in the overall manufacturing sector.

Figure 8 reports the impulse response functions for the four-digit NAICS sub-industries of interest. We estimate that an oil news shock leads to an immediate, large, and statistically significant increase in job destruction for motor vehicle manufacturing and other transportation equipment. Statistically significant increases in job destruction are also observed for motor vehicle parts and for motor vehicle bodies and trailers two quarters after the shock. The rise in job destruction for motor vehicles and related industries is greater and more persistent than for any other industry. As for job creation, while no significant response is evident for motor vehicle parts and other transportation in the first year following the shock, a delayed increase is observed for motor vehicles and motor vehicle bodies and trailers, yet this increase is not enough to offset the rise in job destruction. As a result, net employment growth declines.

The cumulative responses of net employment (NET), gross reallocation (SUM) and excess reallocation (EXC) reported in Table 3 provide a summary of the impact oil news shocks have on the labor dynamics for the sub-industries in transportation equipment. Of note is the significant fourth-quarter cumulative response of job reallocation in motor vehicles (3.14 percentage points), motor vehicle parts (0.34), and other transportation equipment (1.22). These changes are economically significant, as they represent increases of 28.1, 4.0, and 12.3% relative to the historical average reallocation rates for these industries.

5 What factors explain the heterogeneous response of sectoral job creation and destruction to oil news shocks?

Estimation results presented in the previous section reveal a significant degree of heterogeneity in the response of job creation and destruction to oil news shocks. Such heterogeneity is perhaps not surprising as a broad body of empirical literature has documented significant differences in the response of manufacturing sectors to unexpected increases in oil prices. What characteristics drive this variation in industry-level responses? Davis and Haltiwanger (2001) find that the sensitivity of job creation and destruction to oil price shocks is related to the degree of energy intensity, capital intensity, and size of the establishment. Similarly, the work by Kehrig and Ziebarth (2017) suggests that the effect of oil supply shocks is greater in firms with higher capital-energy complementarities. How important are these factors in accounting for the different industry responses to oil news shocks? To answer this question, we exploit the variation in the cumulative job flow responses and inquire whether cross-industry differences in the size of the responses are systematically related to the above-mentioned factors.

5.1 Estimation Strategy and Industry-level Characteristics

As mentioned above, one advantage of using the ASM/CMF data is that we can compute measures of job creation and destruction at a more disaggregated level (four-digit NAICS) than publicly available. This allows us to leverage the variation in responses across industries to explore what sectoral characteristics matter for the transmission of oil news shocks. Another advantage of these data is that they allow us to construct industry-level measures of energy intensity, capital intensity, size, age, and other industry-level characteristics over the period under analysis and then investigate the role of these characteristics in explaining cross-industry variation in impulse responses. To investigate what industry-level characteristics play a role in the transmission of oil news shocks, we proceed in the following manner. First, we compute job creation and destruction rates that span 1980Q3-2016Q4 for 86 four-digit industries. We then reestimate the Proxy-VAR model in equation (4) rotating in the job creation and destruction rates for each of these industries

and compute the cumulative response at different horizons of interest.[10] Similarly, we estimate the cumulative impulse responses of net employment growth, gross reallocation, and excess reallocation. The four-quarter and twelve-quarter cumulative responses, respectively, are employed as a measure of the medium- and long-run responses to an oil news shock.

Second, we construct measures of industry-level characteristics over the 1980Q3-2016Q4 period. Given that cumulative impulse response functions represent the average medium- or long-run responses during the estimation period, all industry-level characteristics are computed by averaging the variables during the sample period. We consider the following variables (or their transformation) as possible determinants of the variation in cross-industry responses to oil news shocks.

Energy-intensity is likely to affect the sensitivity of sectoral output and employment to oil price fluctuations. In fact, the estimation results reported in section 4 suggest that job creation and destruction in sectors that are intensive in the use of energy are more sensitive to news of an oil price increase. Hence, as in Davis and Haltiwanger (2001), we compute a measure of energy-intensity at the establishment level dividing the cost of energy by the total value of shipments. To gain some insight into the differences in job flows across firms with different levels of energy-intensity, we report summary statistics for job creation and destruction by energy deciles in Table 4. This table reveals that the mean and standard deviation of job creation are higher for firms that are less energy-intensive. As for job destruction, while firms in the top three deciles exhibit a somewhat higher mean and volatility than in the bottom three deciles, the differences are small.

Capital-intensity is measured as the cost of capital expenditure per production worker. It is worth noting here that this variable differs from other measures of capital intensity used in the literature in that we do not calculate the stock of physical capital. Instead, we directly use the expenditure reported by the establishment in acquiring and maintaining fixed assets. Thus, our measure captures variation in investment per worker across industries. On the one hand, industries with larger expenditures in energy saving technologies (e.g., fuel efficient motors, enhancement to building envelope, replacement of old inefficient equipment) are likely to face lower increases in production costs when energy prices rise and hence be able to adjust more smoothly to the shock. On the other hand, industries with larger ratios of capital expenditure to labor could have experienced

[10] Due to limits on the disclosure of estimates obtained from the confidential data, we do not report the impulse response estimates at the 4-digit level.

a faster process of automation during the sample period, which could have rendered job flows less responsive to shocks. Given that we cannot differentiate between capital expenditure in energy saving/intensive technologies nor can we account for the degree of automation in each industry, we do not have a prior regarding the relation between capital-intensity and the sensitivity of job flows to oil news shocks. Yet, two interesting patterns can be observed in Table 4 where we summarize the mean and standard deviation of job creation and destruction by capital-intensity deciles. First, average job flows decrease systematically with capital intensity. Second, establishments in the top deciles of capital intensity exhibit less variation in job flows over time. These statistics suggest that capital deepening is correlated with decreased labor market fluidity.

Capital-energy ratio is defined as the ratio of capital expenditures to energy expenditures. Recall that we measure job flows based on the changes in production workers, thus this ratio could be thought of as a proxy for the ratio of capital expenditures to expenditures in unskilled workers. If there are capital-skill and capital-energy complementarities, an increase in the price of energy could result in the substitution of capital for unskilled workers.[11]

Size and age of the establishments in an industry are also characteristics that could influence the dynamic response of businesses and jobs to shocks. For instance, drops in job creation and destruction over the sample have contributed to a decline in employment dynamism, however, this decline in fluidity has not been homogeneous across age and size. Differences in these characteristics may affect the sensitivity of employment to oil news shocks. We define the size of an establishment as the number of production workers reported in a quarter and then take the average size in the industry and over time. To measure an establishment's age we match the ASM/CMF data with the LBD and measure age as the difference between the current year and the first year when the establishment reports positive employment in the LBD.[12] We also consider the percentage of *mature* establishments in the four-digit industry. We define mature as establishments aged nine and above. Table 5 reports the mean and standard deviation of job creation and destruction for establishments in different size brackets split between mature and young firms. As the table illustrates, mature firms account for almost 80% of the employment in manufacturing and, with the exception of the smallest and largest firms, they exhibit larger mean job creation and destruction rates than their

[11] See, e.g. Polgreen and Silos (2009).

[12] Regression results not reported in the paper reveal no statistically significant effect of age^2.

young counterparts.

Establishment growth is defined as the first difference in the log of the number of establishments in the four-digit NAICS industry. The motivation for considering this variable as a possible source of variation in cross-industry responses is twofold. First, differential rates of establishment growth are likely to be linked to differences in market fluidity across industries. In turn, industries with lower labor market fluidity are likely to be less responsive to shocks.[13] Second, the interplay of establishment births and deaths enhances job churn and may lead to productivity gains via creative destruction (see e.g., Aghion and Howitt (1990)). In turn, cross-industry variation in job churning could lead to differences in the dynamic response of job flows to shocks.

5.2 Estimation Results

Table 6 summarizes the key characteristics that explain cross-industry variation in the fourth- and twelfth-quarter cumulative response of job creation, job destruction, net employment growth, and excess job reallocation.[14] Because excess job reallocation is a better measure of the intensity of reallocation activity, we focus on this index instead of the gross job reallocation rate (see e.g., Davis (1998) and Liu (2013)).

We find that variation in energy cost is the main driver of the cross-industry variation. The higher the energy cost, the more likely an industry is to experience job losses in the short- (fourth-quarter) and medium-run (twelve-quarter) and job gains in the medium-run. As a result, employment growth in industries characterized by higher energy costs is more sensitive to oil news shocks. News shocks prompt a more intense process of job reallocation in these industries, as illustrated by the positive and statistically significant coefficients on energy cost in the excess reallocation regressions.

The ratio of capital to energy and the share of mature plants account for statistically significant variation in the cumulative responses of job destruction and excess reallocation, and marginally explain the variation in the response of net employment growth. The magnitude of the cumulative response of job destruction decreases with the capital-energy ratio and is larger in industries with a

[13] See Davis and Haltiwanger (2014) for evidence on the link between labor market fluidity and responsiveness to shocks.

[14] Due to disclosure restrictions, we do not report results for alternatives regression specifications with nonlinear transformations of the regressors or additional explanatory variables when the characteristics are insignificant for all horizons.

higher percentage of mature firms. In addition, the response of excess reallocation is considerably higher in industries with more mature establishments.

Our finding that energy-intensive industries experience more intense job reallocation is in line with previous theoretical and empirical studies that point to energy costs as a key transmission channel of oil price increases to employment and output. However, the finding that the response of job destruction and excess reallocation is greater for industries with a higher percentage of mature firms stands in contrast with Davis and Haltiwanger (2001) who estimate that industries with younger firms are more sensitive to oil price shocks. Differences in the estimation results could stem from two sources: differences in identification strategy and sample period. First, as noted earlier, we identify the effect of oil news shocks using OPEC announcements, whereas Davis and Haltiwanger (2001) rely on short-run restrictions to identify the effect of oil price fluctuations. Hence, our empirical strategy captures an expectations effect that is likely missed by the earlier literature. Second, our sample starts in 1980 and spans until 2016 whereas Davis and Haltiwanger (2001) uses an earlier sample. Thus, changes in the age profile of industries and in the transmission mechanism might explain why we find mature industries to be more responsive to oil news shocks.

Interestingly, we do not find any evidence that the response of excess reallocation is greater for industries where capital intensity is higher. To further investigate this issue, we separate industries by capital intensity quintiles, compute job flows, re-estimate the model in equation (4) and compute the impulse response functions (IRFs) for job creation and job destruction for the lowest and highest capital expenditure quintiles. Figure 9 reveals an important difference in responses, while job destruction for industries in the lowest capital quintile does not change significantly, a significant increase is observed a quarter after the shock for the highest quintile. For both quintiles, job creation increases significantly one year after the shock. As a result, net employment growth increases in the very short-run (two quarters after the shock) for the less capital intensive firms, whereas it decreases for the more capital intensive firms. In summary, while differences in the response by capital quintiles are evident in these impulse response functions, the reshuffling of jobs among firms with different capital-intensity levels takes place in the very short-run. This rapid adjustment to oil news shocks, as well as aggregation by industries, explains why the coefficient of capital intensity is not significant in the cross-industry regressions that use the cumulative responses four quarters after the shock.

6 The Changing Impact of Oil News Shocks

This section investigates whether the responses of job creation and destruction have changed over time. As mentioned earlier, two changes in the US economy prompt this investigation. First, a decline in the fluidity of the US labor market has been widely documented in the literature. In turn, this decline could be related to establishments that became less responsive to shocks over time (Bloom (2009) and Decker et al. (2018)). Second, the transmission of oil price shocks to the labor market may have changed as a result of the shale oil boom in the mid-2000s (see Bjørnland and Skretting (2024)).

To investigate whether job flows have become less responsive to oil news shocks, we re-estimate our Proxy-VARs on expanding subsamples that start from 1980Q3-2000Q1 and end with the full sample. More specifically, we add one observation at a time, reestimate the model, and compute the cumulative impulse response functions for job creation and destruction. We opt to investigate time-variation using this expanding window method on the more aggregated data for several reasons. First, we want to allow for the possibility that the relationship between the oil price and the oil surprises IV varies over time, which current Proxy-TVP-VAR models are not well equipped to do. Second, the long computation time for Proxy-TVP-VARs using the particle Gibbs algorithm proposed by Mumtaz and Petrova (2023) and convergence issues when the number of variables is high –as in our Proxy-SVAR– makes this methodology difficult to apply in our setup. Last but not least, due to limits on research output placed by the Census Bureau on projects using confidential ASM/CMF data, we are only able to use expanding samples of the more aggregated (3-digit NAICS) data.

Figures 10 and 11 plot the cumulative impulse response functions estimated over the expanding subsamples. For the sake of brevity, we focus on the responses for total manufacturing and the industries that consume the most energy from fuel or petroleum (transportation).[15] The top row provides striking evidence of a reduction in the responsiveness of job creation and, especially, destruction to oil news shocks in US manufacturing. The cumulative change in job destruction went from a peak of 5.8% in the earlier samples to less than 1% in the full sample, whereas that of

[15] See `https://eia.gov/energyexplained/use-of-energy/industry.php` and `https://www.eia.gov/todayinenergy/detail.php?id=40752`

job creation decreased from a peak of 3.4% to 0.4%. [16]

The top fuel and petroleum consumers in nondurable industries, such as paper, chemicals, petroleum & coal products, and plastics & rubber, also experienced a decline in the responsiveness of job creation and destruction to oil news shocks. For all of these industries, at peak, the estimated cumulative increase in job destruction declined by an order of magnitude. The response declined from 1.3% to 0.2% for chemicals, 2.6% to 0.6% for petroleum & coal products, 1.4% to 0.3% for paper, and 2.3% to 0.3% for plastics & rubber. Similar decreases were observed for job creation.

We observe a similar pattern for the top fuel consumers in the durable industries. In the early subsamples, the top fuel consumers, transportation equipment, and primary metal products were estimated to immediately reduce job creation by 2.1% and 0.4%, respectively; in contrast, no significant decline is estimated in the later subsamples. In addition, at peak, the cumulative change in job creation decreases by about half for these industries. Regarding job destruction, we estimate maximum cumulative increases of 8.1%, 5.5%, and 4.4% for transportation equipment, primary metal products, and nonmetallic mineral products in the earlier subsamples. In contrast, at the peak, the change triggered by the oil news shock is about 75% smaller in the later subsamples.

We conclude this section by noting that, despite the change in the responsiveness of business-level employment to oil news shocks, the characteristics that explain the cross-industry variability remained roughly unchanged over time. Estimation results, not reported herein due to disclosure limits, indicate that energy intensity and establishment age (or maturity) were key drivers of the heterogeneous response to oil news shocks in the earlier sub-samples. Over time, the ratio of capital to energy expenditures gained importance in accounting for this variation. This evidence suggests that technological change plays a role in the responsiveness of job reallocation to oil news shocks.

7 Conclusions

To investigate the effect of oil news shocks on the creation and destruction of jobs in US manufacturing, we started by computing job flow rates from confidential establishment-level data covering the period between 1980Q3 and 2016Q4. Thus, we first provided a rich characterization of the dynamics of job creation and destruction in US manufacturing. We documented a decline in

[16] Figures for the remaining industries can be found in the Online Appendix.

gross job flows since the late 1980s. Job destruction exhibited greater variation than job creation for almost all the sectors, with the only exception of plastics & rubber. Transportation equipment and food & kindred products averaged the highest employment shares during the sample period.

We then estimated a Proxy-VAR using movements in oil futures prices around OPEC announcements as an instrument. We found a small negative and short-lived decline in net employment growth and an increase in job reallocation for aggregate manufacturing. We demonstrated that this muted aggregate response masks a great degree of heterogeneity across industries. In addition, we showed that the sensitivity of employment growth and job reallocation rose significantly with the degree of energy intensity, the ratio of capital-to-energy spending, the share of mature establishments, and the rate of growth of establishments in the industry.

While our results pointed to important reallocative consequences of oil news shocks, we noted a muted response compared to the larger GDP contraction estimated by Känzig (2021) and the more intense increase in job reallocation estimated by previous studies. We posited that a plausible explanation for the smaller effect of oil news shocks was the decrease in the responsiveness of business-level employment. Another plausible explanation could be related to the role of the shale boom in altering the transmission of oil price shocks. We found evidence of a decline in the responsiveness of sectoral job creation and destruction to oil news shocks since the mid-2000s. We concluded that the timing of the decline in sensitivity supports the relevance of the second explanation.[17]

We close by highlighting three key aspects of the response of sectoral job creation and destruction to oil news shocks. First, there is a great degree of heterogeneity in the sensitivity of manufacturing job flows to oil news across industries. This is true for both the full sample and the subsamples. Secondly, we find that no other sector bears the same degree of reshuffling as the motor vehicle industry. Finally, cross-industry variation is systematically correlated with observable establishment-level characteristics, such as the cost of energy and the percentage of mature firms within the industry. These findings have implications for the implementation of policies aimed at mitigating the economic consequences of oil price shocks and the design of theoretical models. On the one hand, evidence of heterogeneity in the industry-level response to these shocks underscores

[17]In separate work Hanson, Herrera, and Zhao (2024) offer a third explanation: oil news have a muted effect relative to demand-driven shocks. Indeed, they find a much larger impact of demand-driven oil price increases on job reallocation.

the need for policy makers to understand how policies may have radically different effects on employment growth in different sectors. On the other hand, the use of models with representative firms in macroeconomics runs the risk of overlooking the role played by firm heterogeneity in the transmission of oil shocks.

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Table 1: Summary Statistics for Job Creation and Job Destruction, 1980:Q3-2016:Q4

			POS		**NEG**		**NET**	**SUM**	**EXC**
Sector	NAICS	Emp. Share	Mean	S.D.	Mean	S.D.			
Total Manufacturing	300-399	100	4.20	(1.05)	4.38	(1.53)	-0.18	8.58	7.61
Nondurables									
Food & Kindred	311	11.05	6.10	(1.83)	5.90	(2.52)	0.21	12.00	11.17
Beverages & Tobacco	312	1.06	4.80	(1.38)	5.01	(1.47)	-0.21	9.81	8.52
Textiles & Fabrics	313	2.99	3.00	(1.19)	4.03	(1.97)	-1.03	7.03	5.51
Textile Mills	314	1.16	4.36	(1.29)	4.33	(1.68)	0.03	8.69	7.27
Apparel & Accessories	315	4.07	5.39	(2.49)	5.21	(2.72)	0.17	10.60	8.96
Leather & Allied	316	0.72	4.39	(1.87)	5.27	(3.04)	-0.88	9.67	7.80
Paper	322	3.98	2.68	(0.85)	2.94	(1.17)	-0.26	5.62	4.87
Printing Matter	323	3.59	3.85	(1.72)	3.74	(1.84)	0.11	7.58	6.56
Petroleum & Coal	324	0.90	3.32	(1.06)	3.54	(1.30)	-0.22	6.86	5.85
Chemicals	325	5.22	2.98	(0.64)	3.32	(1.09)	-0.34	6.30	5.56
Plastics & Rubber	326	5.84	3.94	(1.30)	3.84	(1.10)	0.10	7.78	6.80
Durables									
Wood Products	321	3.31	4.78	(1.66)	4.50	(2.40)	0.28	9.28	7.52
Nonmetallic Minerals	327	3.14	4.35	(1.11)	4.42	(1.71)	-0.07	8.77	7.58
Primary Metals	331	4.72	2.98	(1.12)	3.65	(1.92)	-0.66	6.63	5.07
Fabricated Metals	332	9.19	4.27	(1.36)	4.33	(1.88)	-0.06	8.60	7.31
Machinery, Non-Electrical	333	7.57	4.24	(1.17)	4.60	(1.69)	-0.36	8.84	7.48
Computer & Electronics	334	7.26	3.94	(1.30)	4.66	(1.94)	-0.72	8.60	7.11
Electrical Equipment	335	3.92	3.59	(1.03)	4.10	(1.52)	-0.50	7.69	6.43
Transportation Equipment	336	13.46	4.42	(1.62)	4.62	(1.92)	-0.20	9.04	7.27
Motor Vehicles	3361	-	5.63	(4.41)	5.55	(4.69)	0.08	11.18	11.10
Motor Vehicle Bodies	3362	-	5.67	(2.11)	5.31	(2.83)	0.36	10.99	8.11
Motor Vehicle Parts	3363	-	4.19	(2.09)	4.44	(2.20)	-0.24	8.63	6.41
Other Transportation Equipment	3369	-	4.81	(2.21)	5.11	2.55)	-0.30	9.93	7.53
Furniture & Fixtures	337	3.46	4.27	(1.39)	4.19	(1.83)	0.08	8.45	7.02
Miscellaneous Other	339	3.40	4.49	(1.05)	4.31	(1.38)	0.18	8.81	7.90

Notes: This table reports the average quarterly rates of job creation (POS, destruction (NEG), net employment (NET), gross reallocation (SUM) and excess reallocation(EXC) for total manufacturing, 3-digit NAICS sectors, and key 4-digit NAICS industries. Employment shares are average annual percent of total manufacturing workers within each 3-digit NAICS industry. Standard deviations are reported in parentheses where applicable.

Table 2: Comparison of Average Labor Flows by Industry

NAICS Description	ASM/CMF (1980Q3:2016Q4) POS	NEG	BED (1992Q2:2016Q4) POS	NEG	DH2001 (1972Q2:1988Q4)) POS	NEG
Total Manufacturing	4.20	4.38	4.1	4.4	5.2	5.5
Nondurables						
Food & Kindred	6.10	5.90	5.4	5.4	9.0	9.0
Beverages & Tobacco	4.80	5.01	5.6	5.3	-	-
Textiles & Fabrics	3.00	4.03	3.4	4.8	-	-
Textile Mills	4.36	4.33	5.3	6.0	3.5	4.1
Apparel & Accessories	5.39	5.21	6.7	8.6	5.4	6.3
Leather & Allied	4.39	5.27	4.6	5.7	4.8	6.1
Paper	2.68	2.94	2.8	3.3	3.6	3.8
Printing Matter	3.85	3.74	4.4	5.0	4.5	4.5
Petroleum & Coal 3.32	3.54	3.9	4.2	4.9	5.3	
Chemicals	2.98	3.32	2.9	3.1	3.6	4.0
Plastics & Rubber	3.94	3.84	3.9	4.0	5.4	5.3
Durables						
Wood Products	4.78	4.50	5.2	5.5	6.6	6.9
Nonmetallic Minerals	4.35	4.42	5.1	5.3	5.3	5.7
Primary Metals	2.98	3.65	2.9	3.5	3.7	4.6
Fabricated Metals	4.27	4.33	4.5	4.6	5.3	5.6
Machinery, Non-Electrical	4.24	4.60	3.8	4.0	4.9	5.3
Computer & Electronics	3.94	4.66	3.2	3.7	-	-
Electrical Equipment	3.59	4.10	3.2	3.7	4.7	4.9
Transportation Equipment	4.42	4.62	3.4	-3.6	5.5	5.7
Furniture & Fixtures	4.27	4.19	4.9	5.4	5.1	5.3
Miscellaneous Other	4.49	4.31	4.6	4.8	6.7	7.1

Notes: Average job creation (POS) and destruction (NEG) rates from the ASM/CMF private data, BED public access data, and Davis and Haltiwanger (2001) reported estimates. - denotes that the industry was not reported for the sample.

Table 3: Cumulative Changes in Labor Flows by Sector

	NET		SUM		EXC	
Quarters after shock	0	4	0	4	0	4
Total Manufacturing	0.02	0.03	0.10	**0.54**	0.08	0.51
Nondurables						
Food & Kindred	0.13	-0.17	-0.03	0.50	-0.15	0.33
Beverages & Tobacco	**0.46**	0.31	-0.04	-0.28	-0.51	-0.60
Textiles & Fabrics	0.06	-0.31	-0.19	0.25	-0.25	-0.06
Textile Mills	0.22	-0.17	-0.16	0.42	-0.38	0.25
Apparel & Accessories	-0.07	-0.02	-0.17	0.69	-0.24	0.67
Leather & Allied	0.18	0.15	**-0.48**	-0.30	**-0.66**	-0.45
Paper	0.00	-0.02	0.13	**0.54**	0.13	**0.52**
Printing Matter	0.17	0.06	**-0.29**	0.22	**-0.46**	-0.49
Petroleum & Coal	0.05	-0.08	**0.34**	**0.76**	0.29	**0.67**
Chemicals	0.15	0.08	0.02	0.06	-0.13	-0.03
Plastics & Rubber	0.15	0.19	-0.14	-0.05	**-0.29**	-0.23
Durables						
Wood Products	**0.49**	0.38	-0.02	0.70	**-0.51**	0.32
Nonmetallic Minerals	-0.12	-0.34	**0.25**	**1.03**	0.13	**0.69**
Primary Metals	-0.23	-0.07	0.12	0.22	-0.11	0.15
Fabricated Metals	-0.11	0.11	-0.02	0.53	-0.13	0.42
Machinery, Non-Electrical	-0.12	0.12	-0.18	-0.14	**-0.29**	-0.26
Computer & Electronics	-0.01	0.11	-0.07	-0.05	-0.08	-0.11
Electrical Equipment	0.05	0.11	-0.18	0.00	-0.24	-0.11
Transportation Equipment	-0.18	-0.38	**0.32**	**1.05**	0.14	0.67
Motor Vehicles	**-1.61**	**-1.61**	**0.75**	**3.14**	**-0.85**	1.53
Motor Vehicle Bodies	-0.34	-1.47	0.21	2.02	-0.13	0.56
Motor Vehicle Parts	0.18	-0.42	0.04	**0.34**	-0.14	-0.08
Other Transportation Equipment	-0.75	-1.23	**0.88**	**1.22**	0.13	-0.01
Furniture & Fixtures	0.12	-0.19	-0.09	**0.51**	-0.21	0.33
Miscellaneous Other	-0.13	0.03	-0.26	0.21	-0.38	**0.18**

Notes: This table reports the cumulative change (measured in percentage points) 0 and 4 quarters after the shock for net employment growth (NET), gross job reallocation (SUM), and excess reallocation (EXC) to a positive innovation in the real oil price. We normalize the oil news shock to 10% increase in oil price. 68% confidence bands are computed via 10,000 bootstrap and significance is denoted in bold.

Table 4: Summary Statistics for Job Creation and Job Destruction by Energy and Capital Decile, 1980:Q3-2016:Q4

		Job Creation		**Job Destruction**	
	Emp. Share (%)	Mean	S.D.	Mean	S.D.
Energy Decile					
0 - 9	10	4.669	(1.269)	4.617	(1.671)
10 - 19	10	4.679	(1.168)	4.759	(1.659)
20 - 29	10	4.459	(1.235)	4.606	(1.996)
30 - 39	10	4.274	(1.350)	4.406	(1.878)
40 - 49	10	4.175	(1.080)	4.389	(1.676)
50 - 59	10	4.040	(1.029)	4.275	(1.488)
60 - 69	10	3.980	(1.093)	4.402	(1.542)
70 - 79	10	3.946	(1.069)	4.404	(1.451)
80 - 89	10	3.760	(0.925)	4.376	(1.374)
90 - 100	10	3.798	(0.956)	4.553	(1.513)
Capital Decile					
0 - 9	10	4.663	(1.833)	5.433	(2.634)
10 - 19	10	4.523	(1.468)	5.047	(2.341)
20 - 29	10	4.406	(1.340)	4.767	(1.953)
30 - 39	10	4.351	(1.248)	4.582	(1.726)
40 - 49	10	4.238	(1.061)	4.511	(1.666)
50 - 59	10	4.136	(1.095)	4.393	(1.630)
60 - 69	10	4.035	(0.955)	4.230	(1.369)
70 - 79	10	3.846	(0.861)	4.123	(1.180)
80 - 89	10	3.820	(0.849)	4.137	(1.221)
90 - 100	10	3.853	(0.921)	4.092	(1.384)

Notes: This table reports the average quarterly job creation and destruction rates for each energy and capital decile. Standard deviations are reported in parentheses.

Table 5: Summary Statistics for Job Creation and Job Destruction by Firm Age-Size Category, 1980:Q3-2016:Q4

		Job Creation		**Job Destruction**	
	Emp. Share (%)	Mean	S.D.	Mean	S.D.
Mature Firms (≥9 Years)					
0 - 19 Workers	2.57	0.872	(0.996)	5.437	(6.851)
20 - 99 Workers	13.87	3.718	(2.162)	4.685	(3.050)
100 - 199 Workers	18.56	4.232	(2.432)	4.196	(2.773)
200 - 399 Workers	16.26	3.188	(1.827)	2.65	(1.790)
400 - 749 Workers	21.57	3.049	(1.834)	2.138	(1.516)
750+ Workers	6.14	0.393	(0.329)	0.248	(0.228)
Young Firms (<9 Years)					
0 - 19 Workers	1.83	1.988	(4.678)	2.681	(6.601)
20 - 99 Workers	4.25	2.394	(2.640)	1.335	(2.268)
100 - 199 Workers	3.00	1.077	(2.148)	0.770	(1.624)
200 - 399 Workers	3.72	1.053	(2.577)	0.679	(1.609)
400 - 749 Workers	2.93	0.648	(1.528)	0.403	(1.013)
750+ Workers	5.31	0.726	(1.826)	0.330	(0.944)

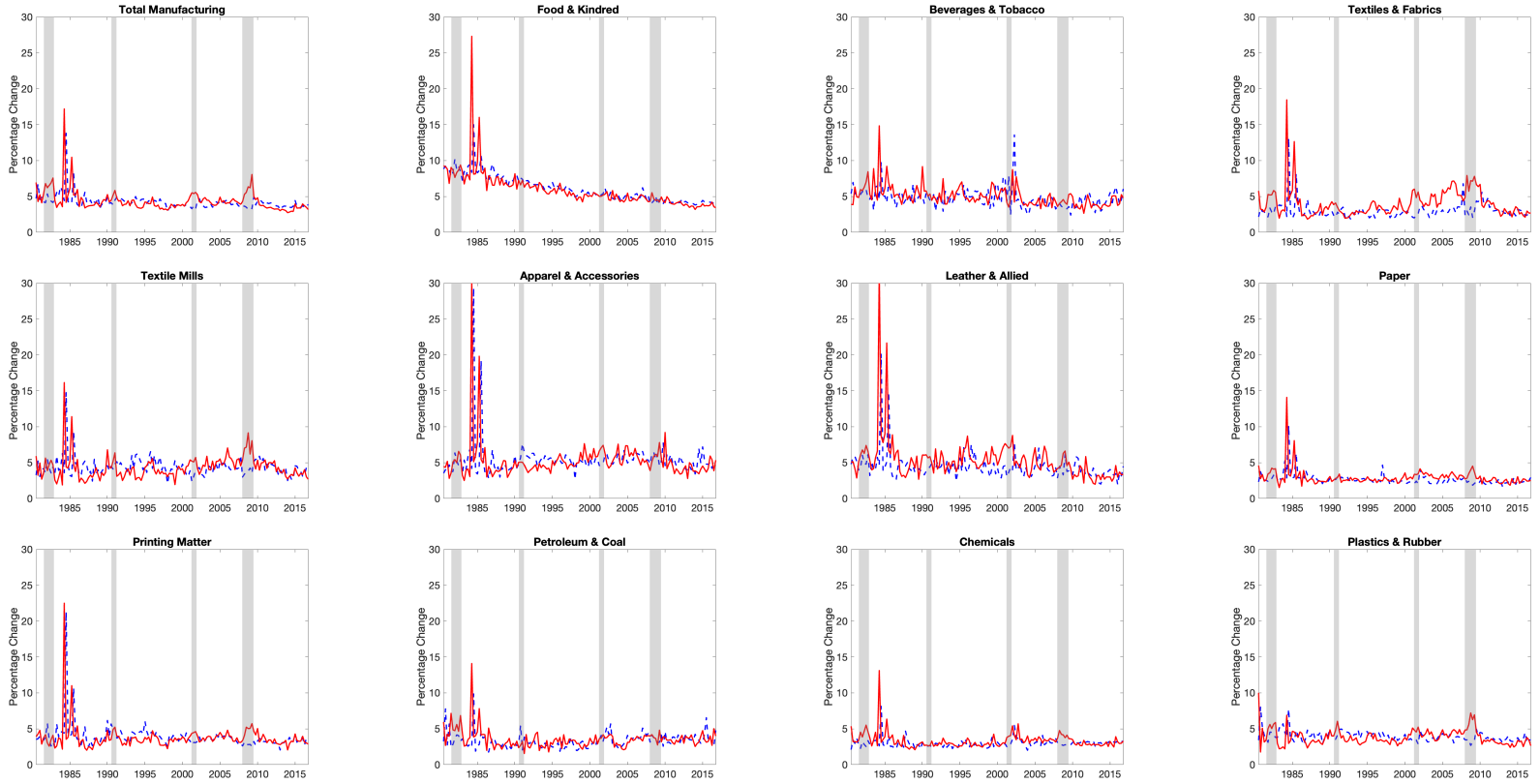
Notes: This table reports the average quarterly job creation and destruction rates for each age-size category. Firms nine years or older are classified as Mature Firms; the remaining firms are classified as young. Employment shares are average annual percent of total manufacturing workers within each age-size category. Standard deviations are reported in parentheses.

Table 6: Relation between Firm-Level Characteristics and the Cumulative Response of Job Creation and Destruction to Oil News Shocks

	Job Creation		**Job Destruction**	
Quarters after the shock	4	12	4	12
Energy-Cost	0.0144	0.0566***	0.0546***	0.0845***
	(0.0087)	(0.0126))	(0.0137)	(0.0174)
Energy-Capital Ratio	-0.000198	-0.000115	-0.000649*	-0.000891**
	(0.000222)	(0.000322)	(0.00035)	(0.000445)
Mature Share	0.0080	0.0158	-0.0033	0.0108
	(0.0067)	(0.0097)	(0.0106)	(0.0134)
Establishment Growth	-0.0021	0.0194	0.0388	0.0737**
	(0.0154)	(0.0223)	(0.0243)	(0.0309)
	Net Employment Growth		**Excess Reallocation**	
Quarters after the shock	4	12	4	12
Energy-Cost	-0.0402**	-0.0280	0.0340**	0.1270***
	(0.0159)	(0.0175)	(0.0168)	(0.0243)
Energy-Capital Ratio	0.000451	0.000775*	-0.000520	-0.000726
	(0.000408)	(0.000448)	(0.000429)	(0.000622)
Mature Share	0.0113	0.0050	0.0235*	0.0570***
	(0.0123)	(0.0135)	(0.0130)	(0.0188)
Establishment Growth	-0.0409	-0.0542*	0.0298	0.0431
	(0.0283)	(0.0310	(0.0298)	(0.0431)

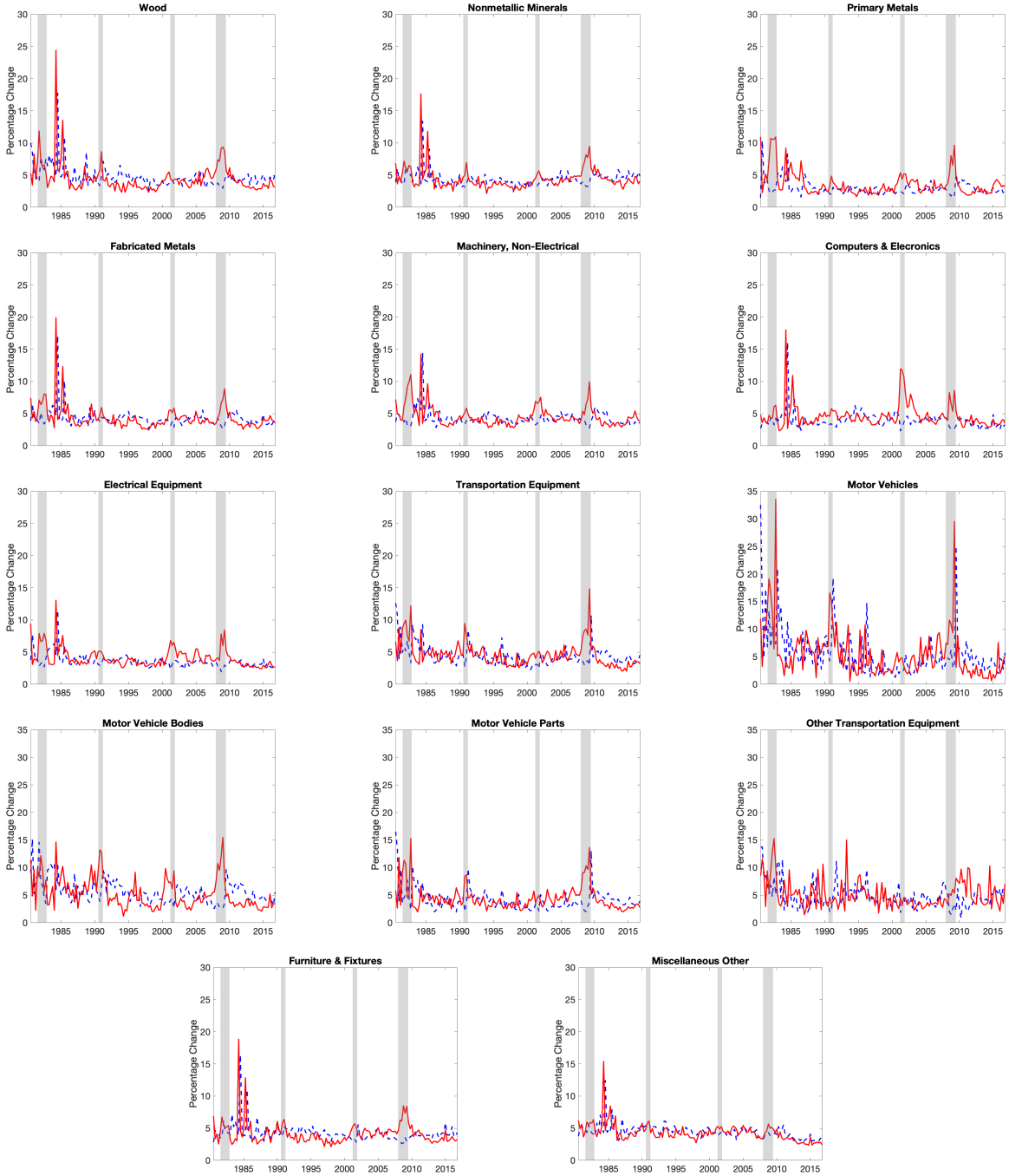
Notes: This table reports results from regressing industry-level cumulative responses to an oil news shock on the indicated industry characteristics. See text for full description of the variables. Robust standard errors are reported in parenthesis. *, **, and *** denote significance level at the 10%, 5% and 1%, respectively.

Figure 1: US Manufacturing Job Creation and Destruction Rates, 1980Q3-2016Q4



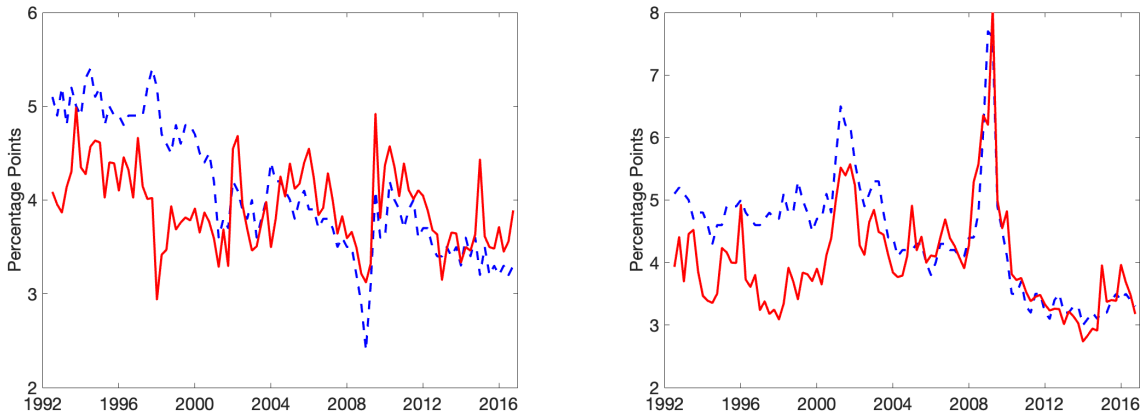
Notes: This figure reports quarterly rates of US manufacturing job creation (POS) and destruction (NEG). Recessionary periods shaded in gray.

Figure 2: US Manufacturing Job Creation and Destruction Rates, 1980Q3-2016Q4 (continued)



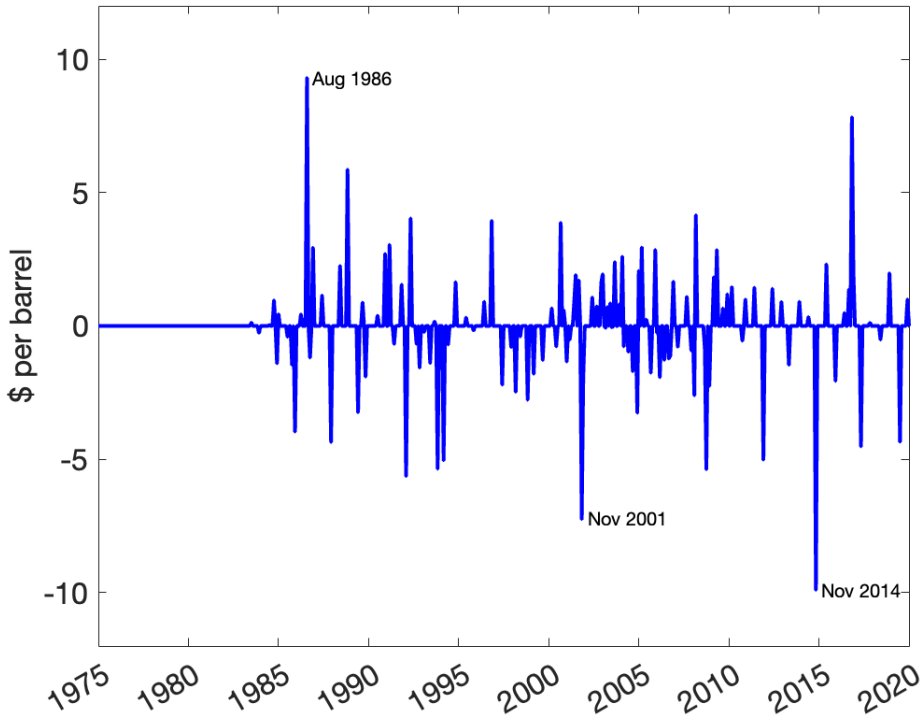
Notes: This figure reports quarterly rates of US manufacturing job creation (POS) and destruction (NEG). Recessionary periods shaded in gray.

Figure 3: Comparison of US Manufacturing Job Creation and Destruction



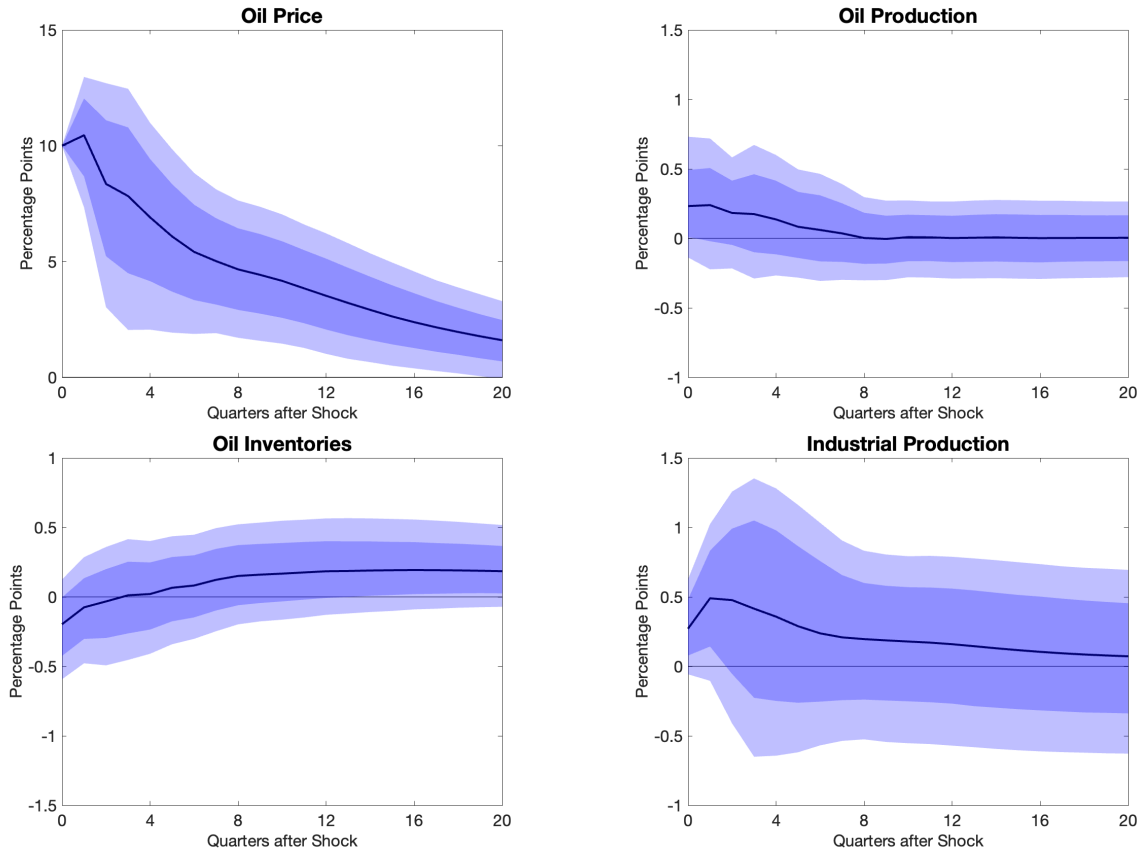
Notes: This figure compares the job creation (left) and destruction (right) rates between the BED (dashed) and ASM/CMF (solid) for the common samples spanning from 1993Q2 to 2016Q4.

Figure 4: Quarterly Oil Price Surprises



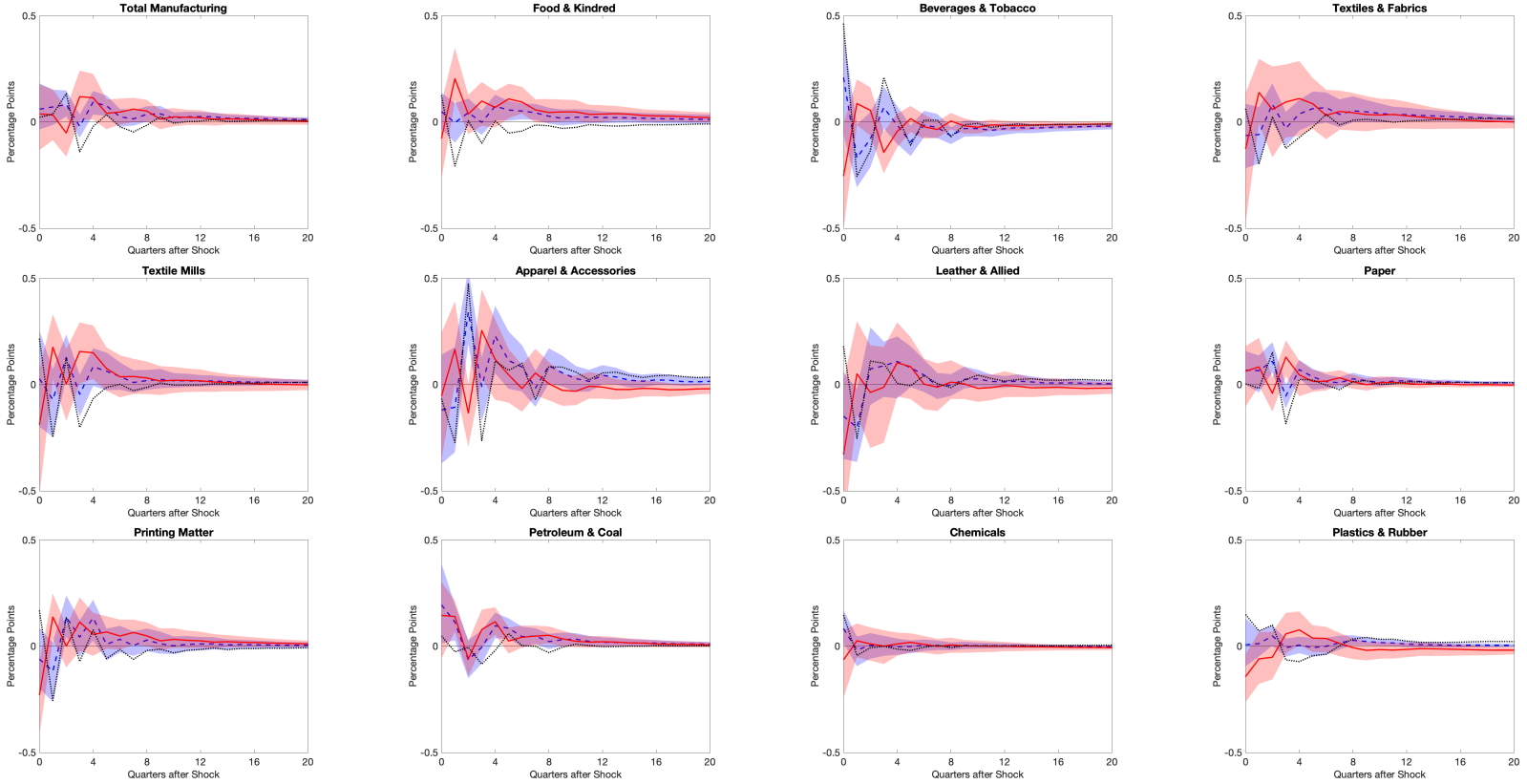
Notes: The figure illustrates the oil news surprise series as the first principle component from changes in oil future prices around OPEC announcements, scaled to match average volatility of underlying price changes.

Figure 5: The Effect of Oil News Shocks on the Oil Market



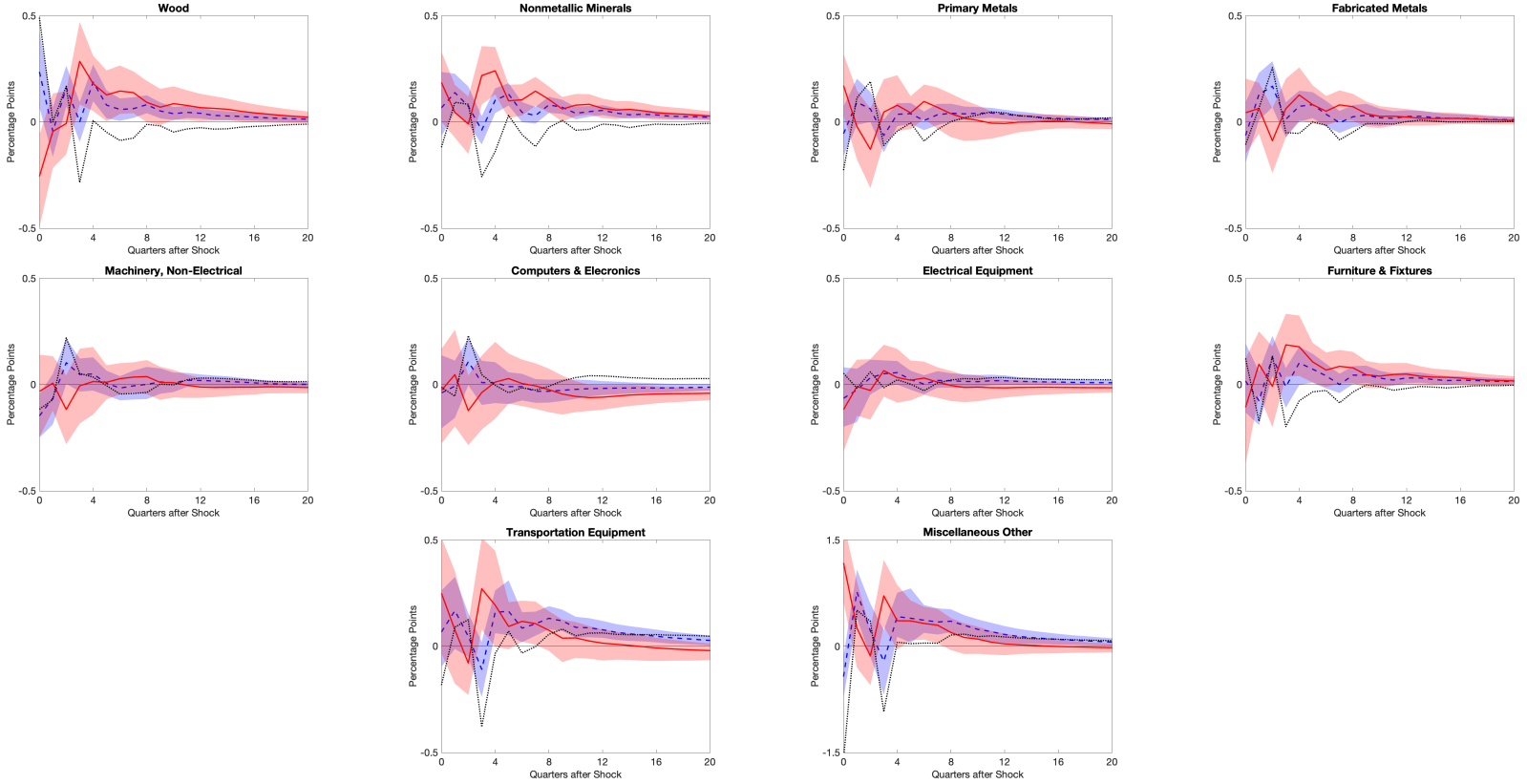
Notes: This figure reports the response of the oil market block to the oil news shock. The shock has been normalized to increase the real oil price by 10 percent on impact. Dark and light shaded areas are 68 and 90 percent confidence bands, respectively, obtained using a moving-block bootstrap with 10,000 repetitions.

Figure 6: The Effect of Oil News Shocks on Job Creation and Destruction



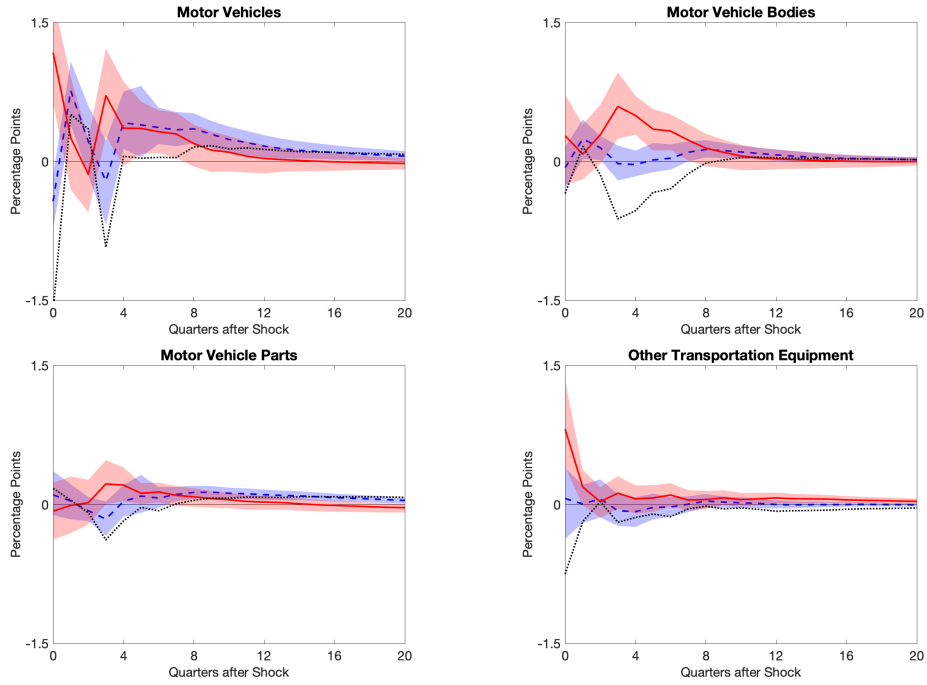
Notes: This figure reports the responses of job creation (dashed blue line) and destruction (solid red line) to an oil news shock. The shock is normalized to increase the real oil price by 10 percent on impact. Shaded areas represent 68 percent confidence bands obtained using a moving-block bootstrap with 10,000 repetitions.

Figure 7: The Effect of Oil News Shocks on Job Creation and Destruction (continued)



Notes: This figure reports the responses of job creation (dashed blue line) and destruction (solid red line) to an oil news shock. The shock is normalized to increase the real oil price by 10 percent on impact. Shaded areas represent 68 percent confidence bands obtained using a moving-block bootstrap with 10,000 repetitions.

Figure 8: The Effect of Oil News Shocks on Job Creation and Destruction - Transportation Equipment Industries

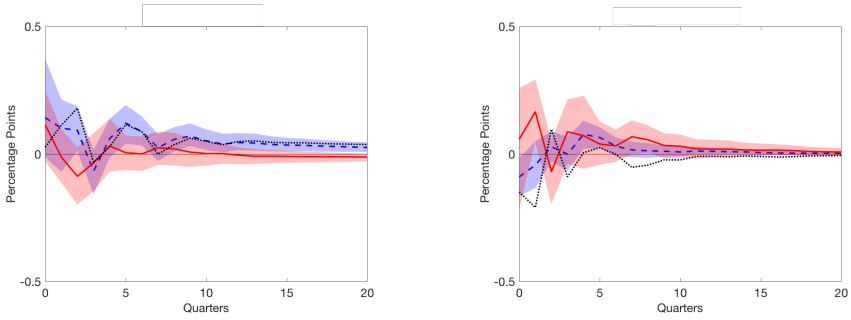


Notes: This figure reports the responses of job creation (dashed blue line) and destruction (solid red line) to an oil news shock. The shock is normalized to increase the real oil price by 10 percent on impact. Shaded areas represent 68 percent confidence bands obtained using a moving-block bootstrap with 10,000 repetitions.

Figure 9: The Effect of Oil News Shocks on Job Creation and Job Destruction - Capital Intensity Quintiles

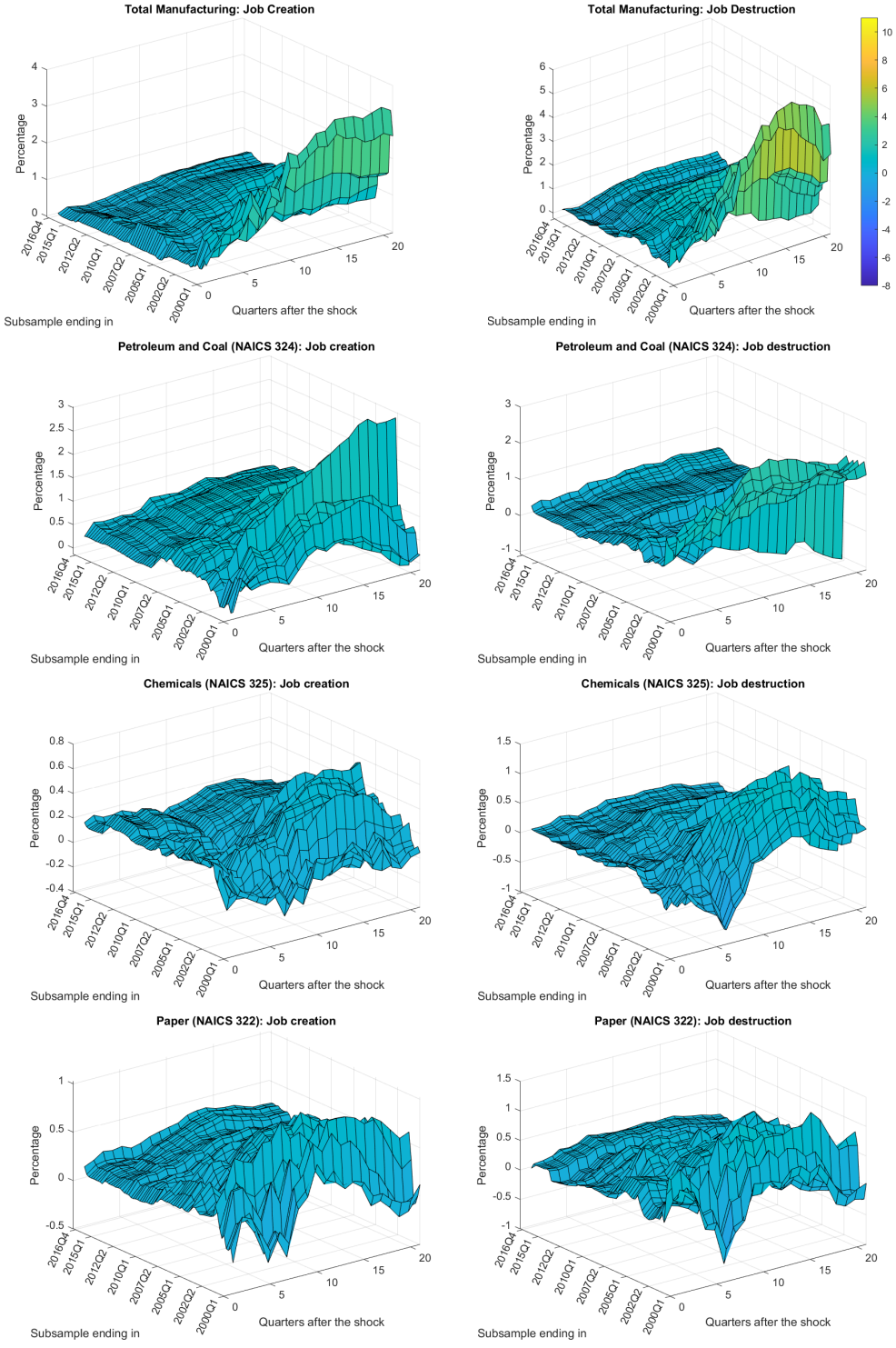
(a) Lowest Capital Intensity Quintile

(b) Highest Capital Intensity Quintile



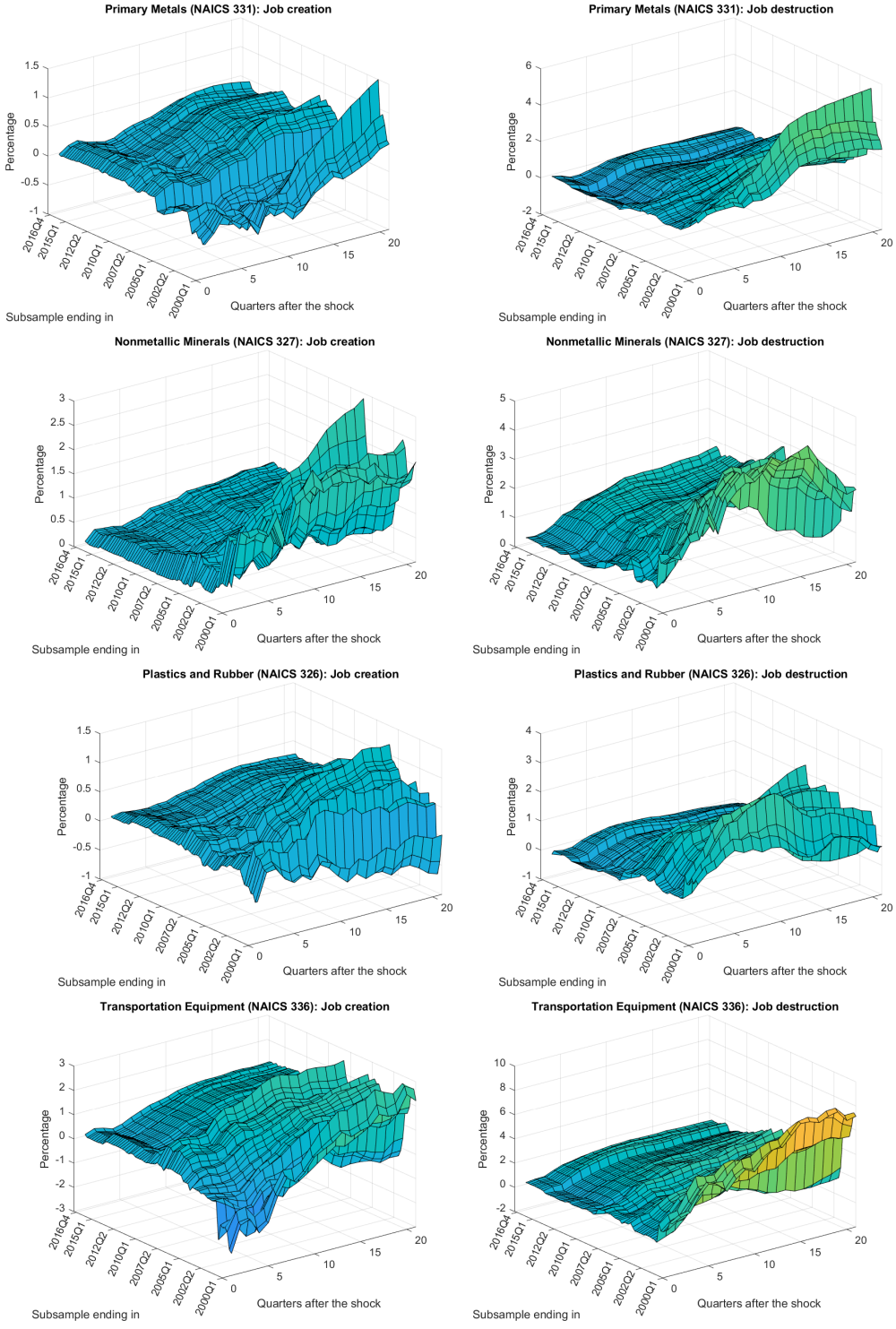
Notes: This figure reports the responses of job creation (dashed blue line) and destruction (solid red line) to an oil news shock. The shock is normalized to increase the real oil price by 10 percent on impact. Shaded areas represent 68 percent confidence bands obtained using a moving-block bootstrap with 10,000 repetitions.

Figure 10: Changes in Responsiveness of Job Flows Over Time



Notes: Surfaces illustrate the changes in the cumulative response of the job creation (left) and destruction (right) rates for subsamples that end on the date noted in the legend.

Figure 11: Changes in Responsiveness of Job Flows Over Time



Notes: Surfaces illustrate the changes in the cumulative response of the job creation (left) and destruction (right) rates for subsamples that end on the date noted in the legend.

A Appendices

A.1 Alternative Model Specifications

We carried out a battery of robustness checks. For the sake of brevity and due to disclosure constraints, we report only the fourth-quarter cumulative responses in Table 7-9 of this Appendix.

Alternative measure of economic activity. We re-estimated the impulse response functions using Kilian's real economic activity (REA) index instead of the world industrial production index. Estimation results reported in the second column of Tables 7-9 show that the sectoral estimation results are qualitatively similar. For total manufacturing, estimates obtained using Kilian's REA suggest a slightly negative impact on net employment growth (-0.015 versus 0.033) and a smaller increase in gross (0.192 versus 0.541) and excess (0.177 versus 0.509) job reallocation. However, the pattern of cross-industry variation is very similar to that observed in the baseline. More specifically, transportation equipment and energy-intensive sectors (e.g., primary metals, nonmetallic minerals) experience the largest declines in net employment and greater reduction in job fluidity.

Alternative measure of oil surprises Kilian (2024) proposes an alternative way to compute the oil surprise IV. We reestimate our Proxy-VARs using the proposed instrument and report the fourth-quarter cumulative response in the third column of Tables 7-10. Using Kilian's measure of surprises we estimate a small cumulative decline in net employment growth for total manufacturing (-0.057) compared to a small increase in the baseline. The pattern of cross-industry variation is also similar to the baseline in this specification. The main difference is a greater degree of variation in the sectoral response of net employment growth and gross reallocation; the standard deviation of the net employment growth and gross reallocation is 2 and 1.5 times that of the baseline, respectively.

Regarding the characteristics that account for cross-industry heterogeneity in the responses, both the baseline model and the Kilian (2024) specification convey a very similar picture. The signs of the coefficients are largely consistent between specifications, with the only exceptions occurring when estimates are not statistically significant. The magnitude and significance of effects vary slightly between models. The main difference is a higher precision of the estimates for the share of mature firms within the industry, which has a small positive effect for almost all jobs when

we use the methodology proposed by Kilian (2024).

Post-1989 data. As discussed above, a possible drawback of employing samples that include the 1980s when identifying oil news shocks using future oil prices is that WTI futures only started trading in 1983 and the market was rather illiquid in the early 1980s. While this concern is alleviated in our study by the fact that we exclude the 1970s data used in Känzig (2021), we reestimate our model using post-1989 (see the fourth column of ables 7-9.). We note that the pattern is similar but the impulse responses are estimated with a lower degree of precision. Using only post-1989 data results in a negative impact on oil news on total manufacturing employment growth, and a larger increase in gross and excess reallocation than estimated with the full sample. The degree of variation across industries is larger for net employment, but smaller for job reallocation. Yet, here –as with the other robustness checks– we find larger impacts on job flows for transportation equipment and the energy-intensive industries than for the less-energy intensive industries.

Pre-2000 data As shown in Section 6, we find evidence of a decline in the magnitude of the responses to oil news shocks since the 2000s. To illustrate how the cumulative responses would have change had we used a shorter sample ending in 199Q4, we reestimate the model using the pre-2000 sample and report the cumulative responses in the fifth column of Tables 7-9. Clearly, of all the specifications, using the pre-200 sample leads to the largest cross-industry variation in the cumulative responses of net employment, gross and net job reallocation. Moreover, this specification leads to the largest estimated decline in employment growth for total manufacturing and for key sectors such as transportation equipment, rubber & plastics and primary metals.

Lag length. The effect of oil news shock on job flows is shorter-lived than that of demand- and supply-driven shocks identified using short-run or sign restrictions. Hence, setting the lag length to $p = 4$ could suffice to capture its dynamics. However, it has been argued that long cycles in crude oil markets would make it necessary to include up to two years of lags (see, e.g. Kilian (2009)). Therefore, we evaluate whether our results are robust to increasing the lag length to eight quarters. Indeed, increasing the lag length leads us to estimate a a larger cumulative drop in employment for transportation equipment, nonmetallic minerals and total manufacturing.Yet, the cross-industry variation shows a very similar pattern to the baseline.

A.2 Details on the Computation of Job Flows

Although the ASM/CMF dataset is restricted access only within the US Census Bureau, for the sake of clarity, we detail some key information that may be relevant to reader understanding and allow reproducibility for researchers with access to the data.

Early-Mid 1980 Values As noted in Section 2, job creation and destruction rates exhibit much greater variability in the 1980s than in any other decade covered in our sample. In particular, we observe large spikes in 1981-1982 for the job creation and destruction rates constructed from the ASM/CMF data that are not present in other data sources like the BDM. However, as Foster et al. (2006) note, "according to the researchers who developed the LBD, these two spikes are the result of a temporary change in the editing methodology of the LBD and should be ignored". Even though we found it reassuring to observe the same spikes in the series constructed by Foster et al. (2006), and we opted to use the series without modifications in our analysis, we investigated the role of imputations in accounting for the spikes. Eliminating the establishments where the number of workers has been imputed and/or otherwise flagged by the U.S. Census does not prevent the job creation and destruction rates from spiking.

Bridging the SIC to NAICS transition. Within the ASM/CMF, establishments were reclassified from the Standard Industrial Classification (SIC) to the North American Industrial Classification System (NAICS) starting in the 1997 CMF and the 1998 ASM. To link establishments across this break, we use a multi-tiered matching system. First, the Census Bureau automatically provides initial linkages for some establishments; when these data were present, we matched accordingly. However, when establishments did not have concrete measures for both SIC and NAICS, we did the following. First, if the establishment identification number occurs both before and after 1998, we define the earlier years industry code as the most recently observed NAICS code post-1998. For establishments that are only observed pre-1998, we use the SIC4-NAICS6 concordance table provided by the Census Bureau. However, only some codes have a one-to-one match. For industries where one SIC code is matched to multiple NAICS codes, we use the employment-weighted concordance table from Schaller and DeCelles (2021) and reallocate the number of production workers into each matched NAICS code according to the industry weight.

Table 7: Comparison of Alternative Models: Net Employment Growth

	NET					
Sector	Baseline	Kilian REA	Kilian Surprise	Post-1989	Pre-2000	8 Lags
Total Manufacturing	0.033	-0.015	-0.057	-0.104	-0.168	-0.015
Nondurables						
Food & Kindred	-0.169	-0.189	-0.127	-0.135	0.742	-0.189
Beverages & Tobacco	0.313	0.590	0.318	0.393	1.781	0.590
Textiles & Fabrics	-0.306	-0.294	0.308	-0.518	0.712	-0.294
Textile Mills	-0.172	-0.927	-0.686	-0.646	-0.806	-0.927
Apparel & Accessories	-0.016	-0.145	-0.005	0.013	1.163	-0.145
Leather & Allied	0.146	0.452	0.685	0.311	0.922	0.452
Paper	-0.02	0.052	0.145	0.022	0.169	0.052
Printing Matter	0.056	0.333	0.125	0.105	0.133	0.333
Petroleum & Coal	-0.085	0.134	0.063	0.357	0.066	0.134
Chemicals	0.084	0.066	-0.023	-0.099	0.356	0.066
Plastics & Rubber	0.186	0.267	0.453	-0.224	-1.725	0.267
Durables						
Wood Products	0.383	0.480	-0.060	0.778	0.288	0.480
Nonmetallic Minerals	-0.338	-0.515	-0.482	-0.217	-0.595	-0.515
Primary Metals	-0.075	0.266	0.136	0.065	-2.683	0.266
Fabricated Metals	0.112	0.114	0.167	0.067	-0.284	0.114
Machinery, Non-Electrical	0.123	-0.251	-0.507	-0.042	-2.626	-0.251
Computer & Electronics	0.208	-0.446	-0.804	-0.510	-0.748	-0.446
Electrical Equipment	0.105	-0.060	0.068	-0.160	-0.594	-0.060
Transportation Equipment	-0.379	-0.787	-0.884	-0.339	-0.618	-0.787
Furniture & Fixtures	-0.185	-0.327	-0.422	-0.220	-0.153	-0.327
Miscellaneous Other	0.027	0.256	0.389	-0.262	0.337	0.256

Notes: This table reports the fourth-quarter cumulative change (measured in percentage points) in net employment (NET) to a positive innovation in the real oil price for each of the main robustness checks described in Appendix A1.

Table 8: Comparison of Alternative Models: Gross Job Reallocation

	SUM					
Sector	Baseline	Kilian REA	Kilian Surprise	Post-1989	Pre-2000	8 Lags
Total Manufacturing	0.541	0.192	-0.218	0.426	-0.362	0.192
Nondurables						
Food & Kindred	0.496	0.853	0.548	0.128	-0.678	0.853
Beverages & Tobacco	-0.283	-0.130	-0.621	-0.296	-2.597	-0.130
Textiles & Fabrics	0.248	-0.043	-0.866	0.163	-1.888	-0.043
Textile Mills	0.422	0.549	0.296	0.476	-1.573	0.549
Apparel & Accessories	0.689	1.350	1.169	0.332	-5.036	1.350
Leather & Allied	-0.302	-0.320	-0.607	-0.413	-2.491	-0.320
Paper	0.544	0.417	0.004	0.340	0.110	0.417
Printing Matter	0.218	-0.298	0.067	0.163	-3.098	-0.298
Petroleum & Coal	0.759	0.769	0.621	0.587	0.797	0.769
Chemicals	0.057	0.267	0.156	0.477	-0.839	0.267
Plastics & Rubber	-0.049	-0.210	-0.508	0.405	0.911	-0.210
Durables						
Wood Products	0.704	-0.051	0.372	0.329	-1.013	-0.051
Nonmetallic Minerals	1.028	0.898	0.644	0.408	0.606	0.898
Primary Metals	0.222	-0.163	-0.433	0.012	1.434	-0.163
Fabricated Metals	0.533	0.012	-0.419	0.453	-1.242	0.012
Machinery, Non-Electrical	-0.141	-0.143	-0.446	0.165	-2.383	-0.143
Computer & Electronics	-0.046	0.079	0.488	0.349	-3.351	0.079
Electrical Equipment	-0.001	-0.136	-0.294	0.250	-1.563	-0.136
Transportation Equipment	1.047	1.081	1.144	0.740	1.100	1.081
Furniture & Fixtures	0.51	0.504	0.328	0.481	-0.491	0.504
Miscellaneous Other	0.208	-0.184	-0.700	0.384	-0.659	-0.184

Notes: This table reports the fourth-quarter cumulative change (measured in percentage points) in total job reallocation (SUM) to a positive innovation in the real oil price for each of the main robustness checks described in Appendix A1.

Table 9: Comparison of Alternative Models: Excess Job Reallocation

	EXC					
Sector	Baseline	Kilian REA	Kilian Surprise	Post-1989	Pre-2000	8 Lags
Total Manufacturing	0.509	0.177	-0.275	0.323	-0.529	0.177
Nondurables						
Food & Kindred	0.327	0.664	0.422	-0.007	-1.420	0.664
Beverages & Tobacco	-0.596	-0.719	-0.939	-0.688	-4.377	-0.719
Textiles & Fabrics	-0.059	-0.337	-1.174	-0.355	-2.599	-0.337
Textile Mills	0.25	-0.378	-0.390	-0.170	-2.379	-0.378
Apparel & Accessories	0.673	1.205	1.164	0.319	-6.199	1.205
Leather & Allied	-0.448	-0.772	-1.291	-0.724	-3.413	-0.772
Paper	0.524	0.365	-0.141	0.318	-0.059	0.365
Printing Matter	0.162	-0.631	-0.058	0.059	-3.231	-0.631
Petroleum & Coal	0.674	0.634	0.558	0.230	0.731	0.634
Chemicals	-0.027	0.201	0.133	0.378	-1.195	0.201
Plastics & Rubber	-0.235	-0.477	-0.961	0.181	-0.813	-0.477
Durables						
Wood Products	0.321	-0.531	0.311	-0.449	-1.301	-0.531
Nonmetallic Minerals	0.691	0.383	0.162	0.190	0.012	0.383
Primary Metals	0.147	-0.429	-0.568	-0.052	-1.249	-0.429
Fabricated Metals	0.421	-0.102	-0.586	0.386	-1.526	-0.102
Machinery, Non-Electrical	-0.263	-0.395	-0.953	0.123	-5.009	-0.395
Computer & Electronics	-0.254	-0.367	-0.316	-0.161	-4.098	-0.367
Electrical Equipment	-0.106	-0.196	-0.362	0.090	-2.157	-0.196
Transportation Equipment	0.667	0.294	0.260	0.402	0.482	0.294
Furniture & Fixtures	0.325	0.178	-0.094	0.260	-0.644	0.178
Miscellaneous Other	0.181	-0.440	-1.089	0.122	-0.996	-0.440

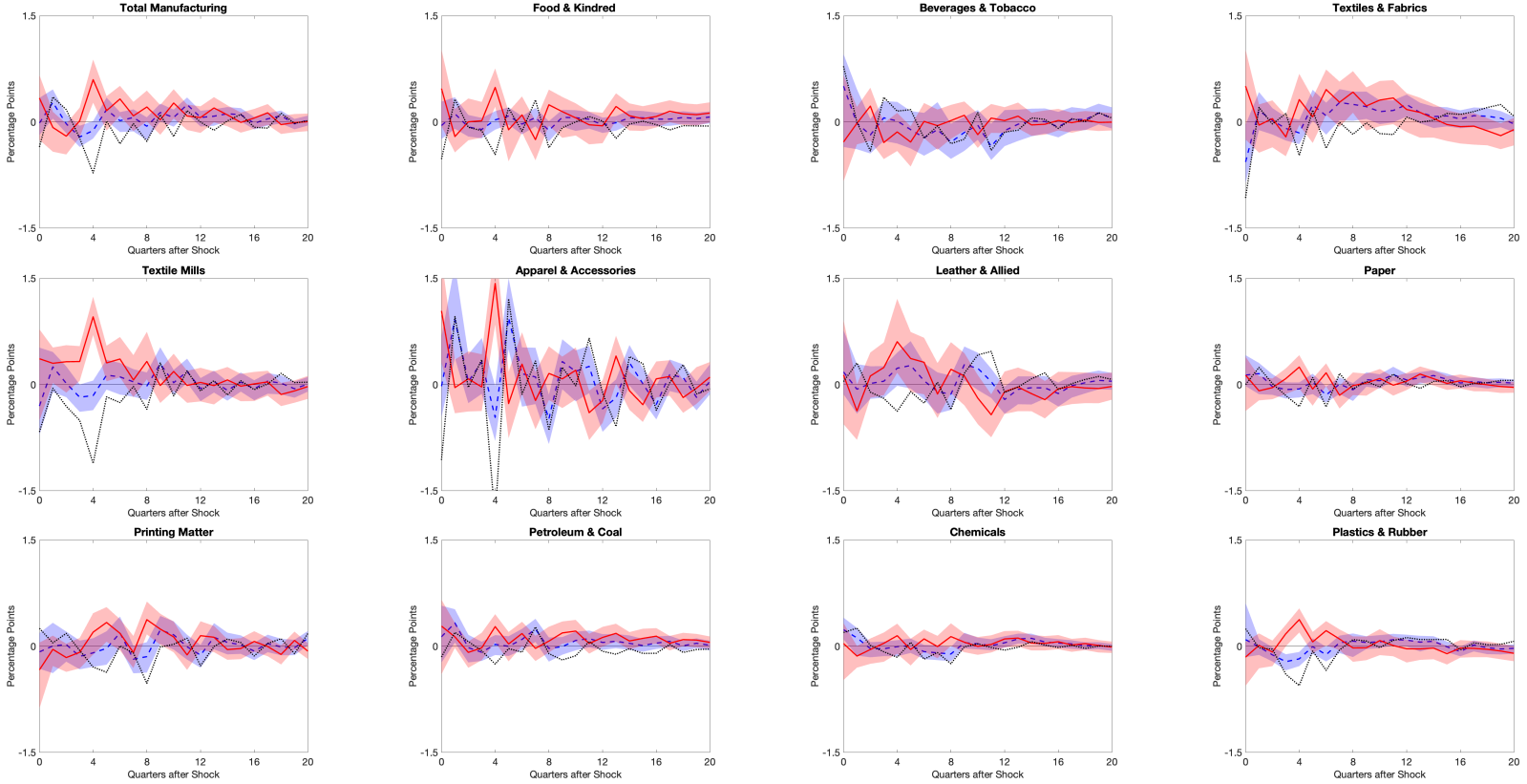
Notes: This table reports the fourth-quarter cumulative change (measured in percentage points) in excess job reallocation (EXC) to a positive innovation in the real oil price for each of the main robustness checks described in Appendix A1.

Table 10: Relation between Firm-Level Characteristics and the Cumulative Response of Job Creation and Destruction to Oil News Shocks - Kilian (2024) Specification

	Job Creation		**Job Destruction**	
Quarters after the shock	4	12	4	12
Energy-Cost	0.0162***	0.0123***	0.0233**	0.00372
	(0.00603)	(0.00416))	(0.00935)	(0.00292)
Energy-Capital Ratio	-0.000239	-0.000038	-0.000458*	0.000424
	(0.000154)	(0.000106)	(0.000239)	(0.000747)
Mature Share	0.0143***	0.0105***	-0.0163**	0.00424*
	(0.00466)	(0.00321)	(0.00722)	(0.00226)
Establishment Growth	-0.0197*	0.0115	0.0247	0.00855
	(0.0107)	(0.0115)	(0.0166)	(0.00518)
	Net Employment Growth		**Excess Reallocation**	
Quarters after the shock	4	12	4	12
Energy-Cost	-0.00714	0.00857*	0.0373***	0.0129**
	(0.0108)	(0.00472)	(0.0118)	(0.00625)
Energy-Capital Ratio	0.00022	0.0000496	-0.000721**	0.000109
	(0.000276)	(0.000121)	(0.000301)	(0.00016)
Mature Share	0.0307***	0.00623*	0.0273***	0.0183**
	(0.00834)	(0.00365)	(0.00909)	(0.00483)
Establishment Growth	-0.04433**	0.00292	-0.0135	0.0224**
	(0.0191)	(0.00838)	(0.0209)	(0.0111)

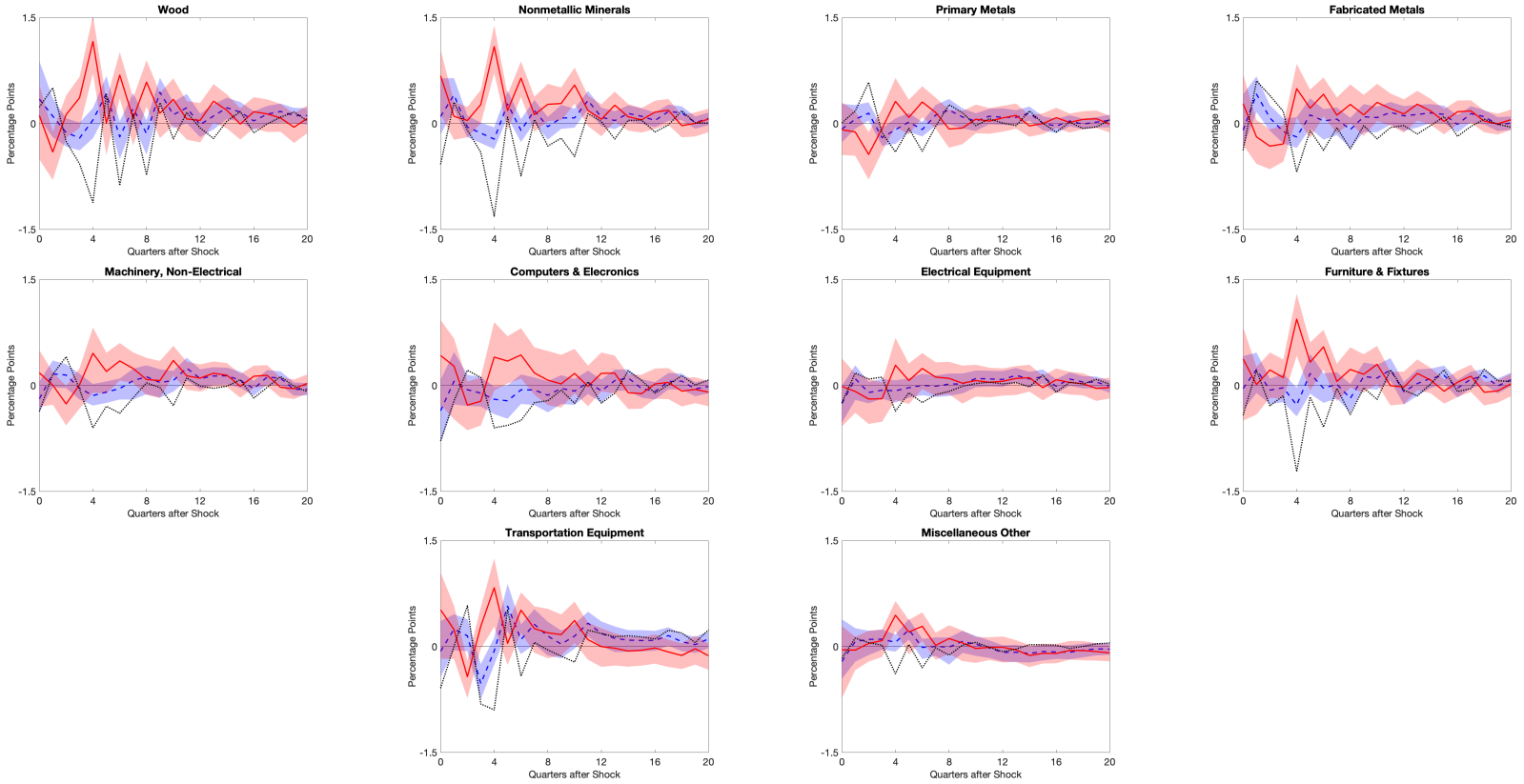
Notes: This table reports results from regressing industry-level cumulative responses to an oil news shock on the indicated industry characteristics, using the Kilian (2024) specifications. See text for full description of the variables. Robust standard errors are reported in parenthesis. *, **, and *** denote significance level at the 10%, 5% and 1%, respectively.

Figure A.1: The Effect of Oil News Shocks on Job Creation and Destruction - Kilian (2024) Methodology



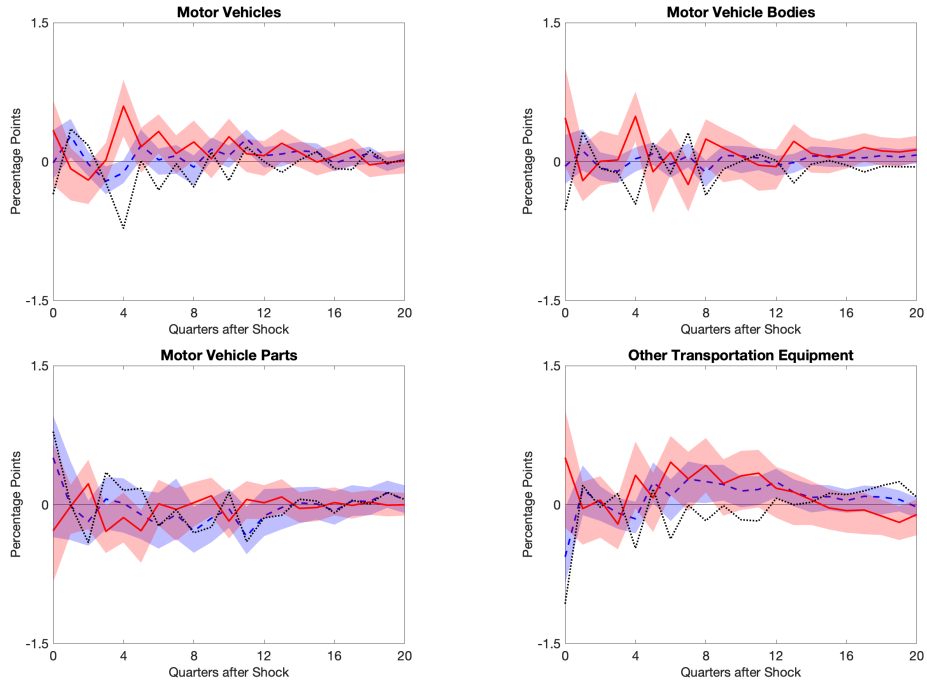
Notes: This figure reports the responses of job creation (dashed blue line) and destruction (solid red line) to an oil news shock. The shock is normalized to increase the real oil price by 10 percent on impact. Shaded areas represent 68 percent confidence bands obtained using a moving-block bootstrap with 10,000 repetitions.

Figure A.2: The Effect of Oil News Shocks on Job Creation and Destruction - Kilian (2024) Methodology (continued)



Notes: This figure reports the responses of job creation (dashed blue line) and destruction (solid red line) to an oil news shock. The shock is normalized to increase the real oil price by 10 percent on impact. Shaded areas represent 68 percent confidence bands obtained using a moving-block bootstrap with 10,000 repetitions.

Figure A.3: The Effect of Oil News Shocks on Job Creation and Destruction - Transportation Equipment Industries - Kilian (2024) Methodology



Notes: This figure reports the responses of job creation (dashed blue line) and destruction (solid red line) to an oil news shock. The shock is normalized to increase the real oil price by 10 percent on impact. Shaded areas represent 68 percent confidence bands obtained using a moving-block bootstrap with 10,000 repetitions.